# electrical-engineering: the Electrical Engineering program, user guide

**electrical-engineering-ai999x 1.6.0**, Rust build: command line, desktop app and browser. **Creator / engineer:** Ibnul Jaif Farabi · **Company:** AI999X · support@ai999x.com

Electrical Engineering covers DC and AC circuits, transients, phasors and power, operational amplifiers, diodes and transistors, digital logic, and control systems (transfer functions, stability, Bode). This guide is about `electrical-engineering`, the model as a self-contained program for Windows and Linux: 92 capabilities in 17 topics, answered on your own computer with no network, no Python and nothing else to install. The same compiled model runs in the AI999X desktop app and in the browser on the model's page (https://www.ai999x.com/models/electrical-engineering-ai999x/); the Python package (`pip install electrical-engineering-ai999x`) has its own guide. Every command output printed here is what the program printed.

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## 1. Get it

Version 1.5.0 of the AI999X desktop app and of the command-line bundle. Checksums for every file: https://downloads.ai999x.com/desktop/1.5.0/SHA256SUMS; the newest version is always under https://downloads.ai999x.com/desktop/latest/. Full instructions: https://www.ai999x.com/desktop-cli/

| what | system | download |
|---|---|---|
| Desktop app: Electrical Engineering | Windows x86-64 | https://downloads.ai999x.com/desktop/1.5.0/ai999x-electrical-engineering-1.5.0-windows-x86_64.zip |
| Desktop app: Electrical Engineering | Linux x86-64 | https://downloads.ai999x.com/desktop/1.5.0/ai999x-electrical-engineering-1.5.0-linux-x86_64.tar.gz |
| Desktop app: all models | Windows x86-64 | https://downloads.ai999x.com/desktop/1.5.0/ai999x-1.5.0-windows-x86_64.zip |
| Desktop app: all models | Linux x86-64 | https://downloads.ai999x.com/desktop/1.5.0/ai999x-1.5.0-linux-x86_64.tar.gz |
| Command line | Windows x86-64 | https://downloads.ai999x.com/desktop/1.5.0/ai999x-cli-1.5.0-windows-x86_64.zip |
| Command line | Linux x86-64 | https://downloads.ai999x.com/desktop/1.5.0/ai999x-cli-1.5.0-linux-x86_64.tar.gz |

### Desktop app

Electrical Engineering in its own app: search, questions in words, input forms, live figures and explanations. Windows: Download ai999x-electrical-engineering-1.5.0-windows-x86_64.zip, right-click it → Extract All, then open the folder and start ai999x-electrical-engineering.exe. If Windows SmartScreen asks, choose More info → Run anyway. Linux:

```
curl -LO https://downloads.ai999x.com/desktop/1.5.0/ai999x-electrical-engineering-1.5.0-linux-x86_64.tar.gz
curl -LO https://downloads.ai999x.com/desktop/1.5.0/SHA256SUMS
sha256sum -c --ignore-missing SHA256SUMS
tar xzf ai999x-electrical-engineering-1.5.0-linux-x86_64.tar.gz
./ai999x-electrical-engineering-1.5.0-linux-x86_64/ai999x-electrical-engineering
```

It opens on this model: type a question or pick a capability, fill in the form and press Compute. The AI999X desktop app with every model works the same way (choose the model first). Windows: Download ai999x-1.5.0-windows-x86_64.zip, right-click it → Extract All, then open the folder and start ai999x.exe. If Windows SmartScreen asks, choose More info → Run anyway. Linux:

```
curl -LO https://downloads.ai999x.com/desktop/1.5.0/ai999x-1.5.0-linux-x86_64.tar.gz
curl -LO https://downloads.ai999x.com/desktop/1.5.0/SHA256SUMS
sha256sum -c --ignore-missing SHA256SUMS
tar xzf ai999x-1.5.0-linux-x86_64.tar.gz
./ai999x-1.5.0-linux-x86_64/ai999x
```

### Command line, Linux

The bundle holds one program per model; `electrical-engineering` is this one. The checksum line should print OK.

```
curl -LO https://downloads.ai999x.com/desktop/1.5.0/ai999x-cli-1.5.0-linux-x86_64.tar.gz
curl -LO https://downloads.ai999x.com/desktop/1.5.0/SHA256SUMS
sha256sum -c --ignore-missing SHA256SUMS
mkdir -p ~/.local/share && tar xzf ai999x-cli-1.5.0-linux-x86_64.tar.gz -C ~/.local/share
echo 'export PATH="$HOME/.local/share/ai999x-cli-1.5.0-linux-x86_64:$PATH"' >> ~/.bashrc &&
source ~/.bashrc
electrical-engineering --version
```

### Command line, Windows (PowerShell)

The fourth line should print True.

```
$F = "ai999x-cli-1.5.0-windows-x86_64.zip"
Invoke-WebRequest "https://downloads.ai999x.com/desktop/1.5.0/$F" -OutFile $F
Invoke-WebRequest "https://downloads.ai999x.com/desktop/1.5.0/SHA256SUMS" -OutFile
SHA256SUMS
(Get-FileHash $F -Algorithm SHA256).Hash.ToLower() -eq ((Select-String -SimpleMatch $F
SHA256SUMS).Line -split ' ')[0]
Expand-Archive $F -DestinationPath "$env:LOCALAPPDATA\AI999X" -Force
$D = "$env:LOCALAPPDATA\AI999X\ai999x-cli-1.5.0-windows-x86_64"; $env:Path += ";$D"
[Environment]::SetEnvironmentVariable("Path", [Environment]::GetEnvironmentVariable("Path",
"User") + ";$D", "User")
electrical-engineering --version
```

### In your browser

Open https://www.ai999x.com/models/electrical-engineering-ai999x/, choose a capability and press Compute in your browser: the same program runs on your device as WebAssembly; nothing is sent anywhere.

### Python

```
pip install electrical-engineering-ai999x
```

## 2. Quick start

### 2.1 Check the install

Prints the model, its version and that this is the Rust build.

```
$ electrical-engineering --version
electrical-engineering-ai999x 1.6.0 (Rust build, ax-engine 1.6.0)
```

### 2.2 Ask in words

Write the question as you would say it: the program reads the quantities and their units, picks the capability and answers.

```
$ electrical-engineering ask "Find the Thevenin equivalent between nodes 2 and 0 of the
circuit: V1 1 0 12; R1 1 2 1000; R2 2 0 2000."
Read as: Thevenin equivalent of a DC circuit between two nodes
Vth = 8 V
Rth = 666.6666667 ohm
  by: Thevenin equivalent of a DC circuit between two nodes  (open-circuit voltage by
modified nodal analysis; resistance by a 1 A test current with the independent sources set
to zero)
  netlist = V1 1 0 12; R1 1 2 1000; R2 2 0 2000.
  node_a = 2
  node_b = 0
  assumes: both nodes in the circuit; no ideal voltage source directly across the
terminals; a unique solution (no floating node)
```

### 2.3 Find a capability

Search the capabilities by what you want to know; the first column is the id to pass to info and solve.

```
$ electrical-engineering search "thevenin equivalent of a dc circuit between two nodes"
thevenin_equivalent                  Thevenin equivalent of a DC circuit between two nodes
(DC circuits / Thevenin and Norton equivalents)
norton_equivalent                    Norton equivalent of a DC circuit between two nodes
(DC circuits / Thevenin and Norton equivalents)
ac_thevenin_equivalent               Thevenin equivalent of an AC circuit between two nodes
(phasors, complex impedance)  (AC circuits / AC Thevenin and Norton equivalents)
ac_norton_equivalent                 Norton equivalent of an AC circuit between two nodes
(phasors, complex impedance)  (AC circuits / AC Thevenin and Norton equivalents)
boolean_equivalence                  Equivalence of two Boolean expressions  (digital logic
/ Boolean algebra)
dc_node_voltages                     Node voltages of a DC resistive circuit from its
netlist  (DC circuits / mesh and nodal analysis)
nodal_voltages                       Node voltages from the nodal equations (Kirchhoff's
current law)  (DC circuits / mesh and nodal analysis)
ac_node_voltages                     Node voltages of an AC circuit from its netlist
(phasor nodal analysis)  (AC circuits / AC network analysis)
… 12 more
```

### 2.4 List the capabilities

Every capability the program answers: its id (what you pass to solve) and its name.

```
$ electrical-engineering list
ac_mesh_currents                         AC mesh currents from the phasor mesh equations
(complex impedances)
ac_node_voltages                         Node voltages of an AC circuit from its netlist
(phasor nodal analysis)
ac_norton_equivalent                     Norton equivalent of an AC circuit between two
nodes (phasors, complex impedance)
ac_thevenin_equivalent                   Thevenin equivalent of an AC circuit between two
nodes (phasors, complex impedance)
adc_conversion                           Analog-to-digital conversion: output code and
resolution
ammeter_shunt                            Ammeter shunt resistance
base_conversion                          Number base conversion (binary, octal, decimal,
hexadecimal)
bjt_divider_bias                         DC operating point of a voltage-divider biased BJT
… 84 more
```

### 2.5 What a capability needs

The equation or method, every input and output with its SI unit. Read this before solve: it tells you the input names.

```
$ electrical-engineering info thevenin_equivalent
Thevenin equivalent of a DC circuit between two nodes
  open-circuit voltage by modified nodal analysis; resistance by a 1 A test current with
the independent sources set to zero
  in   netlist  elements separated by ';': R<name> <node> <node> <ohms>, V<name> <node+>
<node-> <volts>, I<name> <node> <node> <amperes>; node 0 is ground (text) []
  in   node_a   the terminal node (the + side of the equivalent) (text) []
  in   node_b   the reference terminal node (text) []
  out  Vth      Thevenin (open-circuit) voltage, node a relative to node b [V]
  out  Rth      Thevenin resistance [ohm]
```

### 2.6 Solve

Inputs as name=value, each with any unit of the right kind. The answer lists every output, the method used, each input converted to SI, and the conditions the answer assumes.

```
$ electrical-engineering solve thevenin_equivalent node_a=2 node_b=0 "netlist=V1 1 0 12; R1
1 2 1000; R2 2 0 2000."
Vth = 8 V
Rth = 666.6666667 ohm
  by: Thevenin equivalent of a DC circuit between two nodes  (open-circuit voltage by
modified nodal analysis; resistance by a 1 A test current with the independent sources set
to zero)
  netlist = V1 1 0 12; R1 1 2 1000; R2 2 0 2000.
  node_a = 2
  node_b = 0
  assumes: both nodes in the circuit; no ideal voltage source directly across the
terminals; a unique solution (no floating node)
```

### 2.7 Machine-readable answer

--json prints the same answer as one line of JSON (values in SI): for scripts and other programs.

```
$ electrical-engineering solve thevenin_equivalent node_a=2 node_b=0 "netlist=V1 1 0 12; R1
1 2 1000; R2 2 0 2000." --json
{"outcome":"SOLVED","message":"","value":8.0,"unit":"V","solved_for":"Vth","values":
[["Vth",8.0,"V"],["Rth",666.6666666666666,"ohm"]],"target":"Vth","phrase":"Thevenin (open-
circuit) voltage, node a relative to node b","assumptions":["both nodes in the circuit","no
ideal voltage source directly across the terminals","a unique solution (no floating
node)"],"name":"Thevenin equivalent of a DC circuit between two nodes","equation":"open-
circuit voltage by modified nodal analysis; resistance by a 1 A test current with the
independent sources set to zero"}
```

### 2.8 When an input is missing

No number is guessed: the answer names what is missing, shows a complete call, and the program exits with status 1.

```
$ electrical-engineering solve thevenin_equivalent node_a=2 node_b=0
NEEDS_INPUT: thevenin_equivalent needs netlist (elements separated by ';': R<name> <node>
<node> <ohms>, V<name> <node+> <node-> <volts>, I<name> <node> <node> <amperes>; node 0 is
ground). Example: solve('thevenin_equivalent', netlist='V1 1 0 12; R1 1 2 1000; R2 2 0
2000', node_a='2', node_b='0').
```

### 2.9 Many questions at once

One JSON request per line in, one JSON answer per line out, in order: for scripts, notebooks and agents (the program starts once). In PowerShell: Get-Content cases.jsonl | electrical-engineering batch

`cases.jsonl`:

```
{"key": "thevenin_equivalent", "inputs": {"node_a": "2", "node_b": "0", "netlist": "V1 1 0
12; R1 1 2 1000; R2 2 0 2000."}}
{"key": "gain_margin", "inputs": {"num": "[10]", "den": "[1, 6, 5, 0]"}}
```

```
$ electrical-engineering batch < cases.jsonl
{"outcome":"SOLVED","message":"","value":8.0,"unit":"V","solved_for":"Vth","values":
[["Vth",8.0,"V"],["Rth",666.6666666666666,"ohm"]],"target":"Vth","phrase":"Thevenin (open-
circuit) voltage, node a relative to node b","assumptions":["both nodes in the circuit","no
ideal voltage source directly across the terminals","a unique solution (no floating
node)"],"name":"Thevenin equivalent of a DC circuit between two nodes","equation":"open-
circuit voltage by modified nodal analysis; resistance by a 1 A test current with the
independent sources set to zero"}
{"outcome":"SOLVED","message":"","value":3.0,"unit":"","solved_for":"gain_margin","values":
[["gain_margin",3.0,""],["gain_margin_db",9.542425094393248,""],
["phase_crossover",2.23606797749979,"rad/s"]],"target":"gain_margin","phrase":"gain margin
as a ratio","assumptions":["a proper G(s)","exactly one phase crossover with G negative
real"],"name":"Gain margin and phase-crossover frequency","equation":"the phase crossover
as the positive real root of Im(N(jw) conj D(jw)) with Re G < 0 (Aberth iteration, Newton-
polished)"}
```

## 3. Writing inputs

Inputs are written `name=value`, one per argument; `electrical-engineering info <capability>` lists the names. Quote an argument that holds a space or brackets (bash, zsh and PowerShell all accept double quotes). The answer always states each input as it was read, converted to SI, so a misread unit is visible at once.

| kind | example | how it is read |
|---|---|---|
| a number with a unit | `"R=0.128 nm"` | any unit of the same kind; converted to SI |
| a number in SI | `R=1.28e-10` | no unit means the SI unit shown by info |
| a choice | `structure=FCC` | one of the values info lists |
| a list or vector | `"v=[1, 2, 3]" or "v=1, 2, 3"` | brackets keep it a list |
| a matrix | `"A=[[1, 2], [3, 4]]"` | one list per row |
| true / false | `inclusive=true` | for yes/no options |

An equation can be solved for any of its quantities: give the others and add `--target <symbol>`. Without `--target`, the program solves for the quantity the capability is named after.

## 4. Outcomes and exit codes

The program never guesses a number: when it cannot answer, it says why and exits with status 1, so a script can tell an answer from a refusal without reading the text.

| outcome | meaning | exit status |
|---|---|---|
| SOLVED | the answer: every output with its unit, the method, the inputs in SI, the conditions assumed | 0 |
| NEEDS_INPUT | an input is missing; the message names it and shows a complete call | 1 |
| INVALID_INPUT | a value or unit of the wrong kind, an unknown input name, or a value outside the conditions | 1 |
| AMBIGUOUS | the inputs allow two different answers; say which one is meant | 1 |
| OUT_OF_SCOPE | the model has no capability for this, or the result would not be meaningful | 1 |
| (usage) | a command the program does not know: it prints its usage | 2 |

## 5. Scripts and agents: batch and JSON

`electrical-engineering solve ... --json` prints one answer as one line of JSON. `electrical-engineering batch` reads one request per line on standard input and writes one JSON answer per line, in order, starting the program once for any number of requests. A request is `{"key": "<capability>", "inputs": {"name": "value", ...}, "target": null}`. In PowerShell, pipe the file in: `Get-Content cases.jsonl | electrical-engineering batch`. `electrical-engineering catalog` prints every capability with its inputs as JSON, to build forms or tool definitions.

| answer field | meaning |
|---|---|
| outcome | one of the outcomes above |
| message | why there is no answer (empty when SOLVED) |
| value, unit | the main result in SI units |
| values | every output: [symbol, value, unit] (procedures) |
| target | the quantity solved for |
| si | each numeric input converted to SI: [symbol, value] |
| assumptions | the conditions the answer relies on |
| name, equation | the capability and its equation or method |

```
import json, subprocess

reqs = [{"key": "<capability>", "inputs": {"name": "value"}}]
out = subprocess.run(["electrical-engineering", "batch"], input="\n".join(map(json.dumps,
reqs)),
                     capture_output=True, text=True, check=True).stdout
answers = [json.loads(line) for line in out.splitlines()]
```

## 6. Figures

`electrical-engineering figures <capability> name=value ... --out DIR` draws the live figures for those inputs and writes each as an SVG file into DIR (an existing folder); without `--out` the SVG text comes back as JSON. The desktop app and the model page draw the same figures as soon as you press Compute.

## 7. In a web page or Node.js: WebAssembly

The same model is a WebAssembly module, `electrical_engineering_ai999x.wasm` (https://www.ai999x.com/compute/rust/electrical_engineering_ai999x.wasm), with a small loader, `ax-wasm.js` (https://www.ai999x.com/compute/rust/ax-wasm.js). It runs in a browser, a Web Worker or Node.js, with no server.

```
<script src="ax-wasm.js"></script>
<script>
  AXWasm.load("electrical_engineering_ai999x.wasm").then(function (m) {
    var a = m.solve({key: "<capability>", inputs: {name: "value"}});
    console.log(a.outcome, a.values || a.value);
    console.log(m.catalog().capabilities.length + " capabilities");
  });
</script>
```

## 8. Every capability

By topic. For each: what it does, what it needs and gives, when it applies, and a command with what the program printed for it. `electrical-engineering info <id>` shows the same inputs and outputs in the terminal.

### AC circuits (18)

#### AC mesh currents from the phasor mesh equations (complex impedances), `ac_mesh_currents`

Solve the mesh equations Z I = V of an AC circuit in phasor form: the mesh resistance and reactance matrices (Z = R + jX) and the source phasors give every mesh current's magnitude and angle.

$$(\mathbf{R} + j\mathbf{X})\,\mathbf{I} = \mathbf{V}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | mesh resistance matrix in ohms (real part of the mesh impedance matrix) | , |
| `X` | in | mesh reactance matrix in ohms (imaginary part: + wL for an inductor, - 1/(wC) for a capacitor; shared terms with a minus sign) | , |
| `V_mag` | in | rms magnitude of the net source voltage around each mesh, in volts (in the mesh-current direction) | , |
| `V_deg` | in | angle of each mesh's net source voltage, in degrees | , |
| `currents_mag` | out | rms magnitudes of the mesh currents in amperes, in row order | , |
| `currents_deg` | out | angles of the mesh currents in degrees | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `currents_real` | out | real parts of the mesh currents in amperes | , |
| `currents_imag` | out | imaginary parts of the mesh currents in amperes | , |
| `first_mag` | out | magnitude of the first mesh current | A |
| `first_deg` | out | angle of the first mesh current in degrees | , |

**Applies when:** square matrices of one shape; one source phasor per mesh (magnitude >= 0); a non-singular impedance matrix; sinusoidal steady state, one frequency.

```
$ electrical-engineering solve ac_mesh_currents "R=[[10, -4], [-4, 8]]" "X=[[5, -2], [-2,
-3]]" "V_mag=[10, 0]" "V_deg=[0, 0]"
currents_real = [0.984837545126,0.462093862816]
currents_imag = [-0.290252707581,0.274368231047]
currents_mag = [1.02671886344,0.5374092149]
currents_deg = [-16.4213738458,30.6997225508]
first_mag = 1.026718863 A
first_deg = -16.42137385
  by: AC mesh currents from the phasor mesh equations (complex impedances)  (complex
Gaussian elimination with partial pivoting on (R + jX) I = V)
  assumes: square matrices of one shape; one source phasor per mesh (magnitude >= 0); a
non-singular impedance matrix; sinusoidal steady state, one frequency
```

In the worked example, R = [10 -4; -4 8], X = [5 -2; -2 -3], V_mag = [10, 0], V_deg = [0, 0] go through the equation above and give currents_real = [0.9848375451, 0.4620938628], currents_imag = [-0.2902527076, 0.274368231], currents_mag = [1.026718863, 0.5374092149], currents_deg = [-16.42137385, 30.69972255]. It applies only when square matrices of one shape; one source phasor per mesh (magnitude >= 0); a non-singular impedance matrix; sinusoidal steady state, one frequency: outside that it says what is wrong instead of returning a number.

### Balanced three-phase load (wye or delta), `three_phase_load`

Phase voltage, phase and line currents, real, reactive and apparent power and the power factor of a balanced load with impedance R + jX per phase.

$$P = \sqrt{3}\, V_L I_L \cos\phi$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `VL` | in | rms line-to-line voltage | V |
| `R` | in | resistance per phase | ohm |
| `X` | in | reactance per phase (positive inductive, negative capacitive) | ohm |
| `connection` | in | load connection | , |
| `phase_voltage` | out | voltage across each phase of the load | V |
| `phase_current` | out | current in each phase of the load | A |
| `line_current` | out | current in each line | A |
| `P` | out | total real power in W | W |
| `Q` | out | total reactive power in var | W |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `S` | out | total apparent power in VA | W |
| `power_factor` | out | power factor R/\|Z\| | , |

**Applies when:** balanced load and supply; R >= 0.

```
$ electrical-engineering solve three_phase_load connection=wye "VL=400 V" "R=8 ohm" "X=6
ohm"
phase_voltage = 230.9401077 V
phase_current = 23.09401077 A
line_current = 23.09401077 A
P = 12800 W
Q = 9600 W
S = 16000 W
power_factor = 0.8
  by: Balanced three-phase load (wye or delta)  (wye: V_ph = V_L/sqrt(3), I_L = I_ph;
delta: V_ph = V_L, I_L = sqrt(3) I_ph; P = 3 I_ph^2 R)
  VL = 400 V
  R = 8 ohm
  X = 6 ohm
  connection = wye
  assumes: balanced load and supply; R >= 0
```

**What the worked example means:** This calculation determines the electrical characteristics of a balanced three-phase load connected in a wye configuration. Using a line-to-line voltage of 400 V and per-phase impedance defined by resistance 8 ohm and reactance 6 ohm, it computes the phase voltage as 230.94 V, phase current as 23.094 A, and line current as 23.094 A. The resulting total real power is 12800 W, reactive power is 9600 W, apparent power is 16000 W, and the power factor is 0.8. These values indicate how much useful work is performed versus how much energy oscillates between source and load.

**Complex, real, reactive and apparent power from voltage and current phasors, `complex_power`**

S = V I* for rms phasors: real power P, reactive power Q, apparent power |S|, the power factor and whether it lags or leads.

$$S = VI^* = P + jQ, \quad \text{pf} = \frac{P}{|S|}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | rms voltage magnitude | V |
| `V_phase` | in | voltage phase angle in degrees | , |
| `I` | in | rms current magnitude | A |
| `I_phase` | in | current phase angle in degrees | , |
| `P` | out | real (active) power in W | W |
| `Q` | out | reactive power in var (positive: inductive) | W |
| `S` | out | apparent power in VA | W |
| `power_factor` | out | power factor P/\|S\| | , |
| `nature` | out | lagging (inductive), leading (capacitive) or unity | , |

**Applies when:** rms phasors; sinusoidal steady state.

```
$ electrical-engineering solve complex_power "V=230 V" V_phase=0 "I=10 A" I_phase=-30
P = 1991.858429 W
Q = 1150 W
S = 2300 W
power_factor = 0.8660254038
nature = lagging
  by: Complex, real, reactive and apparent power from voltage and current phasors  (S = V
I* (complex conjugate of the current))
  V = 230 V
  V_phase = 0
  I = 10 A
  I_phase = -30
  assumes: rms phasors; sinusoidal steady state
```

**What the worked example means:** This calculation determines the total electrical power flow using voltage and current phasors. Multiplying the voltage phasor by the conjugate of the current phasor gives the complex power 2300 W. The real component represents active power 1991.86 W, which performs useful work, while the imaginary component indicates reactive power 1150 W, associated with energy storage in magnetic or electric fields. The ratio of active to apparent power defines the power factor 0.866025, indicating how efficiently the circuit utilizes the supplied energy.

### Energy stored in an inductor, `inductor_energy`

The magnetic energy L I^2/2 and the flux linkage L I of an inductor carrying current I.

$$W = \tfrac{1}{2} L I^2$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L` | in | inductance | H |
| `I` | in | current | A |
| `energy` | out | stored energy | J |
| `flux_linkage` | out | flux linkage L I in weber-turns | , |

**Applies when:** a linear inductor (no saturation).

```
$ electrical-engineering solve inductor_energy "L=0.05 H" "I=2 A"
energy = 0.1 J
flux_linkage = 0.1
  by: Energy stored in an inductor  (W = L I^2 / 2)
  L = 0.05 H
  I = 2 A
  assumes: a linear inductor (no saturation)
```

**What the worked example means:** This calculation determines the magnetic energy stored in an inductor and its flux linkage based on the given inductance of 0.05 H and current of 2 A. The resulting energy of 0.1 J represents the work done to establish the magnetic field, while the flux linkage of 0.1 indicates the total magnetic flux coupled with the coil turns. These values are critical for analyzing circuit transients and power conversion efficiency.

### First-order RC filter: cutoff, gain and phase, `rc_filter`

The cutoff frequency 1/(2 pi R C) of an RC low-pass or high-pass filter and its gain (ratio and dB) and phase at a frequency.

$$f_c = \frac{1}{2\pi RC}, \quad H_{LP} = \frac{1}{1 + jf/f_c}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | resistance | ohm |
| `C` | in | capacitance | F |
| `f` | in | signal frequency | Hz |
| `filter` | in | first-order low-pass or high-pass | , |
| `fc` | out | cutoff (-3 dB) frequency | Hz |
| `gain` | out | \|Vout/Vin\| at f | , |
| `gain_db` | out | gain in decibels | , |
| `phase_deg` | out | phase of Vout relative to Vin, degrees | , |

**Applies when:** unloaded output; R, C, f > 0.

```
$ electrical-engineering solve rc_filter "R=1000 ohm" "C=1e-07 F" "f=2000 Hz"
filter=lowpass
fc = 1591.549431 Hz
gain = 0.6226769923
gain_db = -4.11474362
phase_deg = -51.48811275
  by: First-order RC filter: cutoff, gain and phase  (H = 1/(1 + j f/fc) (low-pass) or (j
f/fc)/(1 + j f/fc) (high-pass))
  R = 1000 ohm
  C = 1e-07 F
  f = 2000 Hz
  filter = lowpass
  assumes: unloaded output; R, C, f > 0
```

**What the worked example means:** This calculation determines the cutoff frequency, magnitude gain, decibel gain, and phase shift for a first-order RC filter. Using resistance 1000 ohm, capacitance 1e-07 F, and signal frequency 2000 Hz, it finds that the cutoff frequency is 1591.55 Hz. The resulting voltage gain ratio is 0.622677, which corresponds to -4.11474 in decibels, with a phase shift of -51.4881 degrees between output and input.

### Ideal transformer: voltages, currents and reflected resistance, `transformer`

Secondary voltage and current, primary current and the load resistance seen from the primary for an ideal transformer with a resistive load.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}, \quad R_{in} = R_L \left(\frac{N_p}{N_s}\right)^2$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Np` | in | primary turns | , |
| `Ns` | in | secondary turns | , |
| `Vp` | in | primary (rms) voltage | V |
| `RL` | in | load resistance on the secondary | ohm |
| `Vs` | out | secondary voltage | V |
| `Is` | out | secondary (load) current | A |
| `Ip` | out | primary current | A |
| `Rin` | out | load resistance reflected to the primary | ohm |
| `turns_ratio` | out | Np/Ns | , |

**Applies when:** ideal transformer (no losses, no magnetizing current).

```
$ electrical-engineering solve transformer Np=500 Ns=50 "Vp=230 V" "RL=10 ohm"
Vs = 23 V
Is = 2.3 A
Ip = 0.23 A
Rin = 1000 ohm
turns_ratio = 10
  by: Ideal transformer: voltages, currents and reflected resistance  (Vs = Vp Ns/Np, Ip =
Is Ns/Np, Rin = RL (Np/Ns)^2)
  Np = 500
  Ns = 50
  Vp = 230 V
  RL = 10 ohm
  assumes: ideal transformer (no losses, no magnetizing current)
```

**What the worked example means:** This calculation determines the electrical behavior of an ideal transformer based on its turns ratio and connected load. Given a primary voltage of 230 V and a load resistance of 10 ohm, the system computes the resulting secondary voltage as 23 V and the secondary current as 2.3 A. The primary current required to supply this load is found to be 0.23 A, while the resistance seen from the primary side, known as the reflected resistance, is calculated as 1000 ohm. These values illustrate how the transformer scales voltage and impedance between its windings without energy loss.

### Impedance of a parallel RLC circuit, `parallel_rlc_impedance`

The complex impedance of R, L and C in parallel at a frequency: magnitude, phase angle and the two reactances.

$$\frac{1}{Z} = \frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | resistance | ohm |
| `L` | in | inductance | H |
| `C` | in | capacitance | F |
| `f` | in | frequency of the source in Hz | Hz |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `impedance` | out | magnitude of the impedance \|Z\| | ohm |
| `phase_deg` | out | phase angle of Z in degrees (positive: inductive) | , |
| `real` | out | real part of Z | ohm |
| `imag` | out | imaginary part of Z (net reactance) | ohm |
| `XL` | out | inductive reactance wL | ohm |
| `XC` | out | capacitive reactance 1/(wC) | ohm |

**Applies when:** R, L, C, f > 0; sinusoidal steady state.

```
$ electrical-engineering solve parallel_rlc_impedance "R=1000 ohm" "L=0.01 H" "C=1e-06 F"
"f=1000 Hz"
impedance = 103.2622772 ohm
phase_deg = 84.07294187
real = 10.66309789 ohm
imag = 102.7102538 ohm
XL = 62.83185307 ohm
XC = 159.1549431 ohm
  by: Impedance of a parallel RLC circuit  (Y = 1/R + j(wC - 1/(wL)), Z = 1/Y with w = 2 pi
f)
  R = 1000 ohm
  L = 0.01 H
  C = 1e-06 F
  f = 1000 Hz
  assumes: R, L, C, f > 0; sinusoidal steady state
```

**What the worked example means:** This calculation determines the total complex impedance for a resistor, inductor, and capacitor connected in parallel. Combining the reciprocal of resistance with the susceptances of the reactive components shows that the real part of the impedance is 10.6631 ohm and the imaginary net reactance is 102.71 ohm. The resulting magnitude of 103.262 ohm indicates how much the circuit opposes alternating current flow, while the phase angle of 84.0729 shows whether the current leads or lags the voltage.

### Impedance of a series RLC circuit, `series_rlc_impedance`

The complex impedance of R, L and C in series at a frequency: magnitude, phase angle and the two reactances.

$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | resistance | ohm |
| `L` | in | inductance | H |
| `C` | in | capacitance | F |
| `f` | in | frequency of the source in Hz | Hz |
| `impedance` | out | magnitude of the impedance \|Z\| | ohm |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `phase_deg` | out | phase angle of Z in degrees (positive: inductive) | , |
| `real` | out | real part of Z | ohm |
| `imag` | out | imaginary part of Z (net reactance) | ohm |
| `XL` | out | inductive reactance wL | ohm |
| `XC` | out | capacitive reactance 1/(wC) | ohm |

**Applies when:** R >= 0, L > 0, C > 0, f > 0; sinusoidal steady state.

```
$ electrical-engineering solve series_rlc_impedance "R=100 ohm" "L=0.05 H" "C=1e-05 F"
"f=1000 Hz"
impedance = 314.5621512 ohm
phase_deg = 71.46389381
real = 100 ohm
imag = 298.243771 ohm
XL = 314.1592654 ohm
XC = 15.91549431 ohm
  by: Impedance of a series RLC circuit  (Z = R + j(wL - 1/(wC)) with w = 2 pi f)
  R = 100 ohm
  L = 0.05 H
  C = 1e-05 F
  f = 1000 Hz
  assumes: R >= 0, L > 0, C > 0, f > 0; sinusoidal steady state
```

**What the worked example means:** This calculation determines the total opposition to alternating current in a series circuit containing resistance, inductance, and capacitance. Using the source frequency 1000 Hz, it computes the inductive reactance as 314.159 ohm and the capacitive reactance as 15.9155 ohm. The resulting complex impedance has a real part of 100 ohm and an imaginary part of 298.244 ohm, yielding a total magnitude of 314.562 ohm and a phase angle of 71.4639. A positive phase angle indicates the circuit behaves inductively at this frequency.

### Node voltages of an AC circuit from its netlist (phasor nodal analysis), `ac_node_voltages`

Every node-voltage phasor (magnitude, angle, real and imaginary parts), every voltage-source current and the real and reactive power delivered, for a circuit of R, L, C and independent sinusoidal sources of one frequency.

$$\mathbf{Y}(j\omega)\,\mathbf{V} = \mathbf{I}_s, \quad Y_L = \frac{1}{j\omega L},\ Y_C = j\omega C$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `netlist` | in | elements separated by ';': R<name> <node> <node> <ohms>, L<name> <node> <node> <henries>, C<name> <node> <node> <farads>, V<name> <node+> <node-> <rms volts> [<angle deg>], I<name> <node> <node> <rms amperes> [<angle deg>] (flowing from the first node through the source to the second); node 0 is ground | , |
| `f` | in | frequency of the sources | Hz |
| `node_voltages_mag` | out | rms magnitude of every node voltage relative to node 0, in volts | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `node_voltages_deg` | out | angle of every node voltage, in degrees | , |
| `node_voltages_real` | out | real part of every node voltage, in volts | , |
| `node_voltages_imag` | out | imaginary part of every node voltage, in volts | , |
| `source_currents_real` | out | real part of the current through each voltage source from + to -, in amperes | , |
| `source_currents_imag` | out | imaginary part of that current, in amperes | , |
| `P_delivered` | out | real power delivered by the sources (W) | W |
| `Q_delivered` | out | reactive power delivered by the sources, in var (positive: taken by inductance) | , |

**Applies when:** a ground node 0; R, L, C positive; f > 0, one frequency for every source; a unique solution (no floating node, no loop of voltage sources).

```
$ electrical-engineering solve ac_node_voltages "netlist=V1 1 0 10; R1 1 2 3; L1 2 0 0.01."
"f=50 Hz"
node_voltages_mag = {"1":10.0,"2":7.23216748343}
node_voltages_deg = {"1":0.0,"2":43.6792962299}
node_voltages_real = {"1":10.0,"2":5.23042465083}
node_voltages_imag = {"1":0.0,"2":4.9946876259}
source_currents_real = {"V1":-1.58985844972}
source_currents_imag = {"V1":1.6648958753}
P_delivered = 15.8985845 W
Q_delivered = 16.64895875
  by: Node voltages of an AC circuit from its netlist (phasor nodal analysis)  (modified
nodal analysis with complex admittances 1/R, 1/(j w L), j w C, solved by complex Gaussian
elimination)
  f = 50 Hz
  netlist = V1 1 0 10; R1 1 2 3; L1 2 0 0.01.
  assumes: a ground node 0; R, L, C positive; f > 0, one frequency for every source; a
unique solution (no floating node, no loop of voltage sources)
```

In the worked example, netlist = V1 1 0 10; R1 1 2 3; L1 2 0 0.01., f = 50 Hz go through the equation above and give node_voltages_mag = {'1': 10.0, '2': 7.23216748343}, node_voltages_deg = {'1': 0.0, '2': 43.6792962299}, node_voltages_real = {'1': 10.0, '2': 5.23042465083}, node_voltages_imag = {'1': 0.0, '2': 4.9946876259}. It applies only when a ground node 0; R, L, C positive; f > 0, one frequency for every source; a unique solution (no floating node, no loop of voltage sources): outside that it says what is wrong instead of returning a number.

### Norton equivalent of an AC circuit between two nodes (phasors, complex impedance), `ac_norton_equivalent`

The Norton current phasor and impedance seen between two nodes of an AC circuit of R, L, C and independent sinusoidal sources at one frequency.

$$\mathbf{I}_N = \frac{\mathbf{V}_{th}}{Z_{th}}, \quad Z_N = Z_{th}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `netlist` | in | elements separated by ';': R<name> <node> <node> <ohms>, L<name> <node> <node> <henries>, C<name> <node> <node> <farads>, V<name> <node+> <node-> <rms volts> [<angle deg>], I<name> <node> <node> <rms amperes> [<angle deg>] (flowing from the first node through the source to the second); node 0 is ground | , |
| `node_a` | in | the terminal node (the + side of the equivalent) | , |
| `node_b` | in | the reference terminal node | , |
| `f` | in | frequency of the sources | Hz |
| `In_mag` | out | rms magnitude of the Norton (short-circuit) current from node a to node b | A |
| `In_deg` | out | angle of the Norton (short-circuit) current from node a to node b in degrees | , |
| `In_real` | out | real part of the Norton (short-circuit) current from node a to node b | A |
| `In_imag` | out | imaginary part of the Norton (short-circuit) current from node a to node b | A |
| `Zn_real` | out | resistance (real part of the equivalent impedance) | ohm |
| `Zn_imag` | out | reactance (imaginary part of the equivalent impedance, + inductive) | ohm |
| `Zn_mag` | out | magnitude of the equivalent impedance | ohm |
| `Zn_deg` | out | angle of the equivalent impedance in degrees | , |

**Applies when:** both nodes in the circuit; no ideal voltage source (or zero impedance) directly across the terminals; a unique solution (no floating node); f > 0.

```
$ electrical-engineering solve ac_norton_equivalent "netlist=V1 1 0 10; R1 1 2 3; L1 2 0
0.01." node_a=2 node_b=0 "f=50 Hz"
In_real = 3.333333333 A
In_imag = 0 A
In_mag = 3.333333333 A
In_deg = 0
Zn_real = 1.569127395 ohm
Zn_imag = 1.498406288 ohm
Zn_mag = 2.169650245 ohm
Zn_deg = 43.67929623
  by: Norton equivalent of an AC circuit between two nodes (phasors, complex impedance)
(Thevenin voltage and impedance by phasor modified nodal analysis, then I_N = V_th / Z_th)
  f = 50 Hz
  netlist = V1 1 0 10; R1 1 2 3; L1 2 0 0.01.
  node_a = 2
  node_b = 0
  assumes: both nodes in the circuit; no ideal voltage source (or zero impedance) directly
across the terminals; a unique solution (no floating node); f > 0
```

In the worked example, netlist = V1 1 0 10; R1 1 2 3; L1 2 0 0.01., node_a = 2, node_b = 0, f = 50 Hz go through the equation above and give In_real = 3.333333333 A, In_imag = 0 A, In_mag = 3.333333333 A, In_deg = 0. It applies only when both nodes in the circuit; no ideal voltage source (or zero impedance) directly across the terminals; a unique solution (no floating node); f > 0: outside that it says what is wrong instead of returning a number.

### Phasor from polar to rectangular form, `polar_to_rectangular`

The real and imaginary parts of a phasor given as a magnitude and an angle in degrees.

$$M\angle\theta = M\cos\theta + jM\sin\theta$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `magnitude` | in | magnitude (non-negative) | , |
| `angle_deg` | in | angle in degrees | , |
| `real` | out | real part | , |
| `imag` | out | imaginary part | , |

**Applies when:** magnitude >= 0.

```
$ electrical-engineering solve polar_to_rectangular magnitude=10 angle_deg=30
real = 8.660254038
imag = 5
  by: Phasor from polar to rectangular form  (x = M cos(theta), y = M sin(theta))
  magnitude = 10
  angle_deg = 30
  assumes: magnitude >= 0
```

**What the worked example means:** This calculation transforms a phasor defined by its magnitude and phase angle into rectangular coordinates. Using the magnitude 10 and the angle 30, the real component is determined as 8.66025 and the imaginary component as 5. These values represent the horizontal and vertical projections of the phasor vector on the complex plane, which are essential for performing arithmetic operations like addition and subtraction on alternating current quantities.

### Phasor from rectangular to polar form, `rectangular_to_polar`

The magnitude and angle (degrees, in (-180, 180]) of a phasor given as real + j imaginary.

$$x + jy = \sqrt{x^2 + y^2} \angle \operatorname{atan2}(y, x)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `real` | in | real part | , |
| `imag` | in | imaginary part | , |
| `magnitude` | out | magnitude | , |
| `angle_deg` | out | angle in degrees | , |

**Applies when:** not both parts zero.

```
$ electrical-engineering solve rectangular_to_polar real=3 imag=4
magnitude = 5
angle_deg = 53.13010235
  by: Phasor from rectangular to polar form  (M = hypot(x, y), theta = atan2(y, x) (the
quadrant kept))
  real = 3
  imag = 4
  assumes: not both parts zero
```

**What the worked example means:** This calculation converts a complex number from rectangular form to polar form. Given the real component 3 and the imaginary component 4, it determines the vector's length and direction. The resulting magnitude is 5, representing the amplitude of the signal, while the angle is 53.1301, indicating its phase relative to the reference axis.

### Power-factor correction capacitor, `power_factor_correction`

The shunt capacitance (and its reactive power) that raises a lagging load's power factor to a target value.

$$C = \frac{P(\tan\phi_1 - \tan\phi_2)}{2\pi f V^2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `P` | in | real power of the load | W |
| `V` | in | rms supply voltage across the capacitor | V |
| `f` | in | supply frequency | Hz |
| `pf_initial` | in | present (lagging) power factor | , |
| `pf_target` | in | target (lagging) power factor | , |
| `C` | out | correction capacitance | F |
| `C_uF` | out | correction capacitance in microfarads | , |
| `Qc` | out | reactive power the capacitor supplies, in var | W |
| `Q_initial` | out | reactive power before correction, in var | W |
| `Q_final` | out | reactive power after correction, in var | W |

**Applies when:** single-phase, sinusoidal; both power factors lagging; target > present.

```
$ electrical-engineering solve power_factor_correction "P=10000 W" "V=230 V" "f=50 Hz"
pf_initial=0.7 pf_target=0.95
C = 0.000416101396 F
C_uF = 416.101396
Qc = 6915.19956 W
Q_initial = 10202.04061 W
Q_final = 3286.841052 W
  by: Power-factor correction capacitor  (Qc = P (tan acos pf1 - tan acos pf2), C = Qc / (2
pi f V^2))
  P = 10000 W
  V = 230 V
  f = 50 Hz
  pf_initial = 0.7
  pf_target = 0.95
  assumes: single-phase, sinusoidal; both power factors lagging; target > present
```

**What the worked example means:** This calculation determines the shunt capacitor value needed to improve the power factor of an inductive load. Comparing the initial lagging power factor of 0.7 with the target of 0.95 shows that the capacitor must provide 6915.2 W of reactive power. Given the real power of 10000 W and supply voltage of 230 V, the resulting capacitance is 0.000416101 F. This reduction in reactive demand helps lower line losses and improves overall system efficiency.

### Resonance, Q factor and bandwidth of a parallel RLC circuit, `parallel_rlc_resonance`

The resonant frequency, quality factor, half-power bandwidth and half-power frequencies of a parallel RLC circuit.

$$f_0 = \frac{1}{2\pi\sqrt{LC}},\ Q = R\sqrt{\frac{C}{L}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | resistance | ohm |
| `L` | in | inductance | H |
| `C` | in | capacitance | F |
| `f0` | out | resonant frequency | Hz |
| `Q` | out | quality factor | , |
| `bandwidth` | out | half-power bandwidth | Hz |
| `f_low` | out | lower half-power frequency | Hz |
| `f_high` | out | upper half-power frequency | Hz |

**Applies when:** ideal R, L, C.

```
$ electrical-engineering solve parallel_rlc_resonance "R=10 ohm" "L=0.01 H" "C=1e-06 F"
f0 = 1591.549431 Hz
Q = 0.1
bandwidth = 15915.49431 Hz
f_low = 157.5944505 Hz
f_high = 16073.08876 Hz
  by: Resonance, Q factor and bandwidth of a parallel RLC circuit  (Q = R sqrt(C/L),
bandwidth = 1/(2 pi R C))
  R = 10 ohm
  L = 0.01 H
  C = 1e-06 F
  assumes: ideal R, L, C
```

**What the worked example means:** This calculation determines the key frequency response characteristics of an ideal parallel RLC circuit based on its component values. Given a resistance of 10 ohm, an inductance of 0.01 H, and a capacitance of 1e-06 F, the system resonates at 1591.55 Hz. The sharpness of this resonance, known as the quality factor, is calculated to be 0.1, which directly dictates the half-power bandwidth of 15915.5 Hz. Consequently, the frequencies where the power drops to half are found at 157.594 Hz and 16073.1 Hz. These metrics are critical for designing filters and tuned circuits that require specific selectivity.

### Resonance, Q factor and bandwidth of a series RLC circuit, `series_rlc_resonance`

The resonant frequency, quality factor, half-power bandwidth and half-power frequencies of a series RLC circuit.

$$f_0 = \frac{1}{2\pi\sqrt{LC}},\ Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | resistance | ohm |
| `L` | in | inductance | H |
| `C` | in | capacitance | F |
| `f0` | out | resonant frequency | Hz |
| `Q` | out | quality factor | , |
| `bandwidth` | out | half-power bandwidth | Hz |
| `f_low` | out | lower half-power frequency | Hz |
| `f_high` | out | upper half-power frequency | Hz |

**Applies when:** ideal R, L, C.

```
$ electrical-engineering solve series_rlc_resonance "R=10 ohm" "L=0.01 H" "C=1e-06 F"
f0 = 1591.549431 Hz
Q = 10
bandwidth = 159.1549431 Hz
f_low = 1513.960154 Hz
f_high = 1673.115097 Hz
  by: Resonance, Q factor and bandwidth of a series RLC circuit  (Q = (1/R) sqrt(L/C),
bandwidth = R/(2 pi L))
  R = 10 ohm
  L = 0.01 H
  C = 1e-06 F
  assumes: ideal R, L, C
```

**What the worked example means:** This calculation determines the key frequency response characteristics of a series RLC circuit using the input resistance 10 ohm, inductance 0.01 H, and capacitance 1e-06 F. The resonant frequency 1591.55 Hz is where the inductive and capacitive reactances cancel each other out. The quality factor 10 quantifies the sharpness of this resonance, while the half-power bandwidth 159.155 Hz defines the range of frequencies over which the circuit delivers significant power. These values are derived from ideal component assumptions and steady-state sinusoidal operation.

**RMS and average value of a periodic waveform, `rms_waveform`**

The rms value (with DC offset), average, AC rms and crest factor of a sine, square, triangle or sawtooth wave.

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2\,dt}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `shape` | in | waveform shape | , |
| `amplitude` | in | peak amplitude about the average | V |
| `offset` | in | DC offset (average value); 0 when not given (optional) | V |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `rms` | out | rms value including the offset | V |
| `average` | out | average value | V |
| `ac_rms` | out | rms of the AC part | V |
| `crest_factor` | out | peak over rms | , |

**Applies when:** symmetric waveform of the stated shape about its offset.

```
$ electrical-engineering solve rms_waveform "amplitude=5 V" "offset=0 V" shape=square
rms = 5 V
average = 0 V
ac_rms = 5 V
crest_factor = 1
  by: RMS and average value of a periodic waveform  (Vrms = sqrt(offset^2 + k A^2) with k =
1/2 (sine), 1 (square), 1/3 (triangle, sawtooth))
  amplitude = 5 V
  offset = 0 V
  shape = square
  assumes: symmetric waveform of the stated shape about its offset
```

**What the worked example means:** This calculation determines the effective voltage, DC offset, and peak stress for a periodic waveform. Given a shape with amplitude 5 V and offset 0 V, the total root mean square is 5 V, while the AC component alone is 5 V. The average value equals the offset at 0 V. The crest factor, which indicates how spiky the signal is relative to its heating power, results in 1. This helps engineers select components that can handle both the continuous thermal load and transient voltage peaks.

### Sum of sinusoids by phasor addition, `phasor_sum`

The amplitude and phase of the sum of sinusoids of one frequency, A_k cos(wt + phi_k), by adding their phasors.

$$\sum_k A_k \angle \phi_k = \sum_k A_k e^{j\phi_k}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `amplitudes` | in | amplitudes (non-negative) | , |
| `phases_deg` | in | phase angles in degrees | , |
| `amplitude` | out | amplitude of the sum | , |
| `phase_deg` | out | phase of the sum in degrees | , |
| `real` | out | real part of the resultant phasor | , |
| `imag` | out | imaginary part of the resultant phasor | , |

**Applies when:** one phase per amplitude; one common frequency.

```
$ electrical-engineering solve phasor_sum "amplitudes=[3, 4, 2]" "phases_deg=[0, 90, 180]"
amplitude = 4.123105626
phase_deg = 75.96375653
real = 1
imag = 4
  by: Sum of sinusoids by phasor addition  (complex addition of the phasors A_k e^{j
phi_k})
  assumes: one phase per amplitude; one common frequency
```

**What the worked example means:** This calculation determines the single sinusoid that results from adding multiple sinusoids of the same frequency. Converting each input amplitude [3, 4, 2] and phase angle [0, 90, 180] into complex phasors, summing them, and converting back gives the final amplitude 4.12311 and phase 75.9638. The real part of this resultant vector is 1 and the imaginary part is 4, representing the combined effect in the time domain.

**Thevenin equivalent of an AC circuit between two nodes (phasors, complex impedance), `ac_thevenin_equivalent`**

The Thevenin voltage phasor and impedance Zth = R + jX seen between two nodes of an AC circuit of R, L, C and independent sinusoidal sources at one frequency.

$$\mathbf{V}_{th} = \mathbf{V}_a - \mathbf{V}_b\big|_{open}, \quad Z_{th} = \frac{\mathbf{V}_a - \mathbf{V}_b}{1\,\mathrm{A}}\Big|_{sources\ off}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `netlist` | in | elements separated by ';': R<name> <node> <node> <ohms>, L<name> <node> <node> <henries>, C<name> <node> <node> <farads>, V<name> <node+> <node-> <rms volts> [<angle deg>], I<name> <node> <node> <rms amperes> [<angle deg>] (flowing from the first node through the source to the second); node 0 is ground | , |
| `node_a` | in | the terminal node (the + side of the equivalent) | , |
| `node_b` | in | the reference terminal node | , |
| `f` | in | frequency of the sources | Hz |
| `Vth_mag` | out | rms magnitude of the Thevenin (open-circuit) voltage, node a relative to node b | V |
| `Vth_deg` | out | angle of the Thevenin (open-circuit) voltage, node a relative to node b in degrees | , |
| `Vth_real` | out | real part of the Thevenin (open-circuit) voltage, node a relative to node b | V |
| `Vth_imag` | out | imaginary part of the Thevenin (open-circuit) voltage, node a relative to node b | V |
| `Zth_real` | out | resistance (real part of the equivalent impedance) | ohm |
| `Zth_imag` | out | reactance (imaginary part of the equivalent impedance, + inductive) | ohm |
| `Zth_mag` | out | magnitude of the equivalent impedance | ohm |
| `Zth_deg` | out | angle of the equivalent impedance in degrees | , |

**Applies when:** both nodes in the circuit; no ideal voltage source (or zero impedance) directly across the terminals; a unique solution (no floating node); f > 0.

```
$ electrical-engineering solve ac_thevenin_equivalent "netlist=V1 1 0 10; R1 1 2 3; L1 2 0
0.01." node_a=2 node_b=0 "f=50 Hz"
Vth_real = 5.230424651 V
Vth_imag = 4.994687626 V
Vth_mag = 7.232167483 V
Vth_deg = 43.67929623
Zth_real = 1.569127395 ohm
Zth_imag = 1.498406288 ohm
Zth_mag = 2.169650245 ohm
Zth_deg = 43.67929623
  by: Thevenin equivalent of an AC circuit between two nodes (phasors, complex impedance)
(open-circuit voltage by phasor modified nodal analysis; impedance by a 1 A test current
with the independent sources set to zero)
  f = 50 Hz
  netlist = V1 1 0 10; R1 1 2 3; L1 2 0 0.01.
  node_a = 2
  node_b = 0
  assumes: both nodes in the circuit; no ideal voltage source (or zero impedance) directly
across the terminals; a unique solution (no floating node); f > 0
```

In the worked example, netlist = V1 1 0 10; R1 1 2 3; L1 2 0 0.01., node_a = 2, node_b = 0, f = 50 Hz go through the equation above and give Vth_real = 5.230424651 V, Vth_imag = 4.994687626 V, Vth_mag = 7.232167483 V, Vth_deg = 43.67929623. It applies only when both nodes in the circuit; no ideal voltage source (or zero impedance) directly across the terminals; a unique solution (no floating node); f > 0: outside that it says what is wrong instead of returning a number.

## Alternating voltages and currents (2)

### Sinusoidal waveform v = Vm sin(wt + phi): frequency, period, phase, value at a time, form and peak factors, `sinusoid_values`

For v = Vm sin(wt + phi): the frequency, period, phase angle in degrees and whether it leads or lags Vm sin wt, the time shift, the value at a time t, the rms and half-cycle average values and the form and peak factors.

$$v = V_m \sin(\omega t + \phi), \quad f = \frac{\omega}{2\pi}, \quad V_{rms} = \frac{V_m}{\sqrt{2}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Vm` | in | amplitude (peak value) | V |
| `w` | in | angular frequency omega | rad/s |
| `phi` | in | phase angle in radians (positive: leads Vm sin wt) | , |
| `t` | in | time at which to evaluate v | s |
| `v` | out | instantaneous value at t | V |
| `f` | out | frequency w/(2 pi) | Hz |
| `T` | out | period | s |
| `phase_deg` | out | phase angle in degrees | , |
| `relation` | out | leads, lags or in phase (relative to Vm sin wt) | , |
| `time_shift` | out | time shift phi/w (positive: leads) | s |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `rms` | out | rms value Vm/sqrt(2) | V |
| `average` | out | average over a half cycle, 2 Vm/pi | V |
| `form_factor` | out | form factor rms/average | , |
| `peak_factor` | out | peak factor Vm/rms | , |

**Applies when:** a pure sinusoid; |phi| <= 2 pi rad.

```
$ electrical-engineering solve sinusoid_values "Vm=200 V" "w=314.159 rad/s" phi=0 "t=0.0025
s"
v = 141.4212624 V
f = 49.99995777 Hz
T = 0.02000001689 s
phase_deg = 0
relation = in phase
time_shift = 0 s
rms = 141.4213562 V
average = 127.3239545 V
form_factor = 1.110720735
peak_factor = 1.414213562
  by: Sinusoidal waveform v = Vm sin(wt + phi): frequency, period, phase, value at a time,
form and peak factors  (v = Vm sin(wt + phi); f = w/(2 pi); T = 1/f; Vrms = Vm/sqrt 2; Vav
= 2 Vm/pi)
  Vm = 200 V
  w = 314.159 rad/s
  phi = 0
  t = 0.0025 s
  assumes: a pure sinusoid; |phi| <= 2 pi rad
```

In the worked example, Vm = 200 V, w = 314.159 rad/s, phi = 0, t = 0.0025 s go through the equation above and give v = 141.4212624 V, f = 49.99995777 Hz, T = 0.02000001689 s, phase_deg = 0. It applies only when a pure sinusoid; |phi| <= 2 pi rad: outside that it says what is wrong instead of returning a number.

### Time at which a sinusoid first reaches a value, `sinusoid_time_to_value`

The first time t >= 0 at which v = Vm sin(wt + phi) equals a given value, the period, and whether v is rising or falling there.

$$\omega t + \phi = \sin^{-1}\frac{v}{V_m} \text{ or } \pi - \sin^{-1}\frac{v}{V_m}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Vm` | in | amplitude (peak value) | V |
| `w` | in | angular frequency omega | rad/s |
| `phi` | in | phase angle in radians (positive: leads Vm sin wt) | , |
| `v` | in | value to reach (\|v\| <= Vm) | V |
| `t` | out | first time at which v is reached | s |
| `T` | out | period | s |
| `direction` | out | rising, falling or peak at that time | , |

**Applies when:** |v| <= Vm; |phi| <= 2 pi rad; t counted from 0.

```
$ electrical-engineering solve sinusoid_time_to_value "Vm=100 V" "w=314.159 rad/s" phi=0
"v=50 V"
t = 0.001666668074 s
T = 0.02000001689 s
direction = rising
  by: Time at which a sinusoid first reaches a value  (wt + phi = asin(v/Vm) or pi -
asin(v/Vm), reduced to the first period; the smaller time)
  Vm = 100 V
  w = 314.159 rad/s
  phi = 0
  v = 50 V
  assumes: |v| <= Vm; |phi| <= 2 pi rad; t counted from 0
```

In the worked example, Vm = 100 V, w = 314.159 rad/s, phi = 0, v = 50 V go through the equation above and give t = 0.001666668074 s, T = 0.02000001689 s, direction = rising. It applies only when |v| <= Vm; |phi| <= 2 pi rad; t counted from 0: outside that it says what is wrong instead of returning a number.

## Basic electrical quantities (4)

### Cells with internal resistance: series and parallel batteries on a load, `cell_battery`

The combined emf and internal resistance of n identical cells in series or in parallel, the load current, the terminal voltage, the lost volts and the load power.

$$I = \frac{E_b}{r_b + R_L}, \quad V = E_b - I r_b$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `E` | in | emf of one cell | V |
| `r` | in | internal resistance of one cell | ohm |
| `n` | in | number of identical cells | , |
| `connection` | in | how the cells are connected | , |
| `RL` | in | load resistance | ohm |
| `emf_total` | out | emf of the battery | V |
| `r_total` | out | internal resistance of the battery | ohm |
| `I` | out | load current | A |
| `V_terminal` | out | terminal voltage (across the load) | V |
| `lost_volts` | out | voltage lost inside the battery, I r | V |
| `P_load` | out | power delivered to the load | W |

**Applies when:** identical cells; E, r, RL > 0; n a whole number.

```
$ electrical-engineering solve cell_battery "E=1.5 V" "r=0.5 ohm" n=4 "RL=10 ohm"
connection=series
emf_total = 6 V
r_total = 2 ohm
I = 0.5 A
V_terminal = 5 V
lost_volts = 1 V
P_load = 2.5 W
  by: Cells with internal resistance: series and parallel batteries on a load  (series: E_b
= n E, r_b = n r; parallel: E_b = E, r_b = r/n; I = E_b/(r_b + R_L), V = I R_L)
  E = 1.5 V
  r = 0.5 ohm
  n = 4
  RL = 10 ohm
  connection = series
  assumes: identical cells; E, r, RL > 0; n a whole number
```

In the worked example, E = 1.5 V, r = 0.5 ohm, n = 4, RL = 10 ohm, connection = series go through the equation above and give emf_total = 6 V, r_total = 2 ohm, I = 0.5 A, V_terminal = 5 V. It applies only when identical cells; E, r, RL > 0; n a whole number: outside that it says what is wrong instead of returning a number.

**Electrical energy and its cost, `electrical_energy_cost`**

The energy a load of power P uses in a time t, in joules and in kilowatt-hours, and its cost at a tariff per kWh.

$$W = Pt, \quad \text{cost} = \frac{Pt}{3.6 \times 10^6\ \text{J/kWh}} \times \text{tariff}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `P` | in | power of the load | W |
| `t` | in | running time (h and min accepted in questions) | s |
| `tariff` | in | price of one kWh (any currency) | , |
| `energy` | out | energy used, in joules | J |
| `energy_kWh` | out | energy used, in kilowatt-hours | , |
| `cost` | out | cost of the energy, in the tariff's currency | , |

**Applies when:** constant power; P > 0, t > 0, tariff >= 0.

```
$ electrical-engineering solve electrical_energy_cost "P=2000 W" "t=18000 s" tariff=0.15
energy = 36000000 J
energy_kWh = 10
cost = 1.5
  by: Electrical energy and its cost  (W = P t; kWh = W / 3.6e6; cost = kWh x tariff)
  P = 2000 W
  t = 18000 s
  tariff = 0.15
  assumes: constant power; P > 0, t > 0, tariff >= 0
```

In the worked example, P = 2000 W, t = 18000 s, tariff = 0.15 go through the equation above and give energy = 36000000 J, energy_kWh = 10, cost = 1.5. It applies only when constant power; P > 0, t > 0, tariff >= 0: outside that it says what is wrong instead of returning a number.

**Resistance at a new temperature (temperature coefficient),** `resistance_at_temperature`

The resistance of a conductor at a temperature T2 from its resistance R1 at T1 and its temperature coefficient referred to T1, and the coefficient referred to T2.

$$R_2 = R_1 \left[1 + \alpha_1 (T_2 - T_1)\right]$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R1` | in | resistance at the reference temperature T1 | ohm |
| `alpha1` | in | temperature coefficient of resistance referred to T1, per K (= per degree C) | , |
| `T1` | in | reference temperature (degrees C accepted in questions) | K |
| `T2` | in | new temperature | K |
| `R2` | out | resistance at T2 | ohm |
| `alpha2` | out | temperature coefficient referred to T2, per K | , |
| `change` | out | change of resistance R2 - R1 | ohm |

**Applies when:** a linear R(T) over the range; R1 > 0, temperatures above absolute zero.

```
$ electrical-engineering solve resistance_at_temperature "R1=50 ohm" alpha1=0.004
"T1=293.15 K" "T2=353.15 K"
R2 = 62 ohm
alpha2 = 0.003225806452
change = 12 ohm
  by: Resistance at a new temperature (temperature coefficient)  (R2 = R1 (1 + alpha1 (T2 -
T1)); alpha2 = alpha1 / (1 + alpha1 (T2 - T1)))
  R1 = 50 ohm
  alpha1 = 0.004
  T1 = 293.15 K
  T2 = 353.15 K
  assumes: a linear R(T) over the range; R1 > 0, temperatures above absolute zero
```

In the worked example, R1 = 50 ohm, alpha1 = 0.004, T1 = 293.15 K, T2 = 353.15 K go through the equation above and give R2 = 62 ohm, alpha2 = 0.003225806452, change = 12 ohm. It applies only when a linear R(T) over the range; R1 > 0, temperatures above absolute zero: outside that it says what is wrong instead of returning a number.

**Temperature coefficient of resistance from two readings,** `temperature_coefficient_from_resistances`

The temperature coefficient of resistance referred to each of two temperatures, from the resistances measured at them.

$$\alpha_1 = \frac{R_2 - R_1}{R_1 (T_2 - T_1)}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R1` | in | resistance at T1 | ohm |
| `T1` | in | first temperature (degrees C accepted in questions) | K |
| `R2` | in | resistance at T2 | ohm |
| `T2` | in | second temperature | K |
| `alpha1` | out | temperature coefficient referred to T1, per K | , |
| `alpha2` | out | temperature coefficient referred to T2, per K | , |
| `slope` | out | change of resistance per kelvin, ohm/K | , |

**Applies when:** a linear R(T) between the readings; T1 != T2.

```
$ electrical-engineering solve temperature_coefficient_from_resistances "R1=100 ohm"
"T1=293.15 K" "R2=120 ohm" "T2=343.15 K"
alpha1 = 0.004
alpha2 = 0.003333333333
slope = 0.4
  by: Temperature coefficient of resistance from two readings  (alpha1 = (R2 - R1) / (R1
(T2 - T1)))
  R1 = 100 ohm
  T1 = 293.15 K
  R2 = 120 ohm
  T2 = 343.15 K
  assumes: a linear R(T) between the readings; T1 != T2
```

In the worked example, R1 = 100 ohm, T1 = 293.15 K, R2 = 120 ohm, T2 = 343.15 K go through the equation above and give alpha1 = 0.004, alpha2 = 0.003333333333, slope = 0.4. It applies only when a linear R(T) between the readings; T1 != T2: outside that it says what is wrong instead of returning a number.

## Capacitors (1)

### Parallel-plate capacitor: capacitance, charge, field, flux density and energy, `parallel_plate_capacitor`

The capacitance of a capacitor of n parallel plates with a dielectric, and its charge, electric field strength, electric flux density and stored energy at a voltage V.

$$C = \frac{\varepsilon_0 \varepsilon_r A (n-1)}{d}, \quad D = \varepsilon_0 \varepsilon_r E, \quad W = \tfrac{1}{2} C V^2$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | voltage across the capacitor | V |
| `d` | in | separation of adjacent plates (dielectric thickness) | m |
| `A` | in | overlapping area of one plate | m^2 |
| `eps_r` | in | relative permittivity of the dielectric (1 for air) | , |
| `n` | in | number of plates (2 when not given) (optional) | , |
| `C` | out | capacitance | F |
| `Q` | out | charge | C |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `E` | out | electric field strength V/d | N/C |
| `D` | out | electric flux density in C/m^2 | , |
| `energy` | out | stored energy | J |

**Applies when:** uniform field (plate size much larger than d); eps_r >= 1, n >= 2 plates.

```
$ electrical-engineering solve parallel_plate_capacitor "V=200 V" "d=0.0001 m" "A=0.0004
m^2" eps_r=5
C = 1.770837564e-10 F
Q = 3.541675127e-08 C
E = 2000000 N/C
D = 8.854187818e-05
energy = 3.541675127e-06 J
  by: Parallel-plate capacitor: capacitance, charge, field, flux density and energy  (C =
eps0 eps_r A (n - 1)/d; Q = CV; E = V/d; D = eps0 eps_r E; W = CV^2/2)
  V = 200 V
  d = 0.0001 m
  A = 0.0004 m^2
  eps_r = 5
  assumes: uniform field (plate size much larger than d); eps_r >= 1, n >= 2 plates
```

In the worked example, V = 200 V, d = 0.0001 m, A = 0.0004 m^2, eps_r = 5 go through the equation above and give C = 1.770837564e-10 F, Q = 3.541675127e-08 C, E = 2000000 N/C, D = 8.854187818e-05. It applies only when uniform field (plate size much larger than d); eps_r >= 1, n >= 2 plates: outside that it says what is wrong instead of returning a number.

## Control systems (11)

### Closed-loop poles with unity negative feedback, `closed_loop_poles`

The poles of K*G(s)/(1 + K*G(s)): the roots of den(s) + K*num(s), and whether the closed loop is stable.

$$1 + K\,G(s) = 0 \iff \mathrm{den}(s) + K\,\mathrm{num}(s) = 0$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `num` | in | numerator coefficients of the open-loop G(s), highest power of s first | , |
| `den` | in | denominator coefficients of G(s), highest power of s first | , |
| `K` | in | loop gain multiplying G(s) (default 1) (optional) | , |
| `poles` | out | closed-loop poles (real first by real part, conjugate pairs together) | , |
| `max_real_part` | out | largest real part of the poles (negative when stable) | , |
| `stable` | out | true when every pole has a negative real part | , |
| `characteristic` | out | closed-loop characteristic polynomial den + K num | , |

**Applies when:** a proper G(s); unity negative feedback.

```
$ electrical-engineering solve closed_loop_poles "num=[2]" "den=[1, 3, 0]"
poles = [-2.0,-1.0000000000000002]
max_real_part = -1
stable = true
characteristic = [1.0,3.0,2.0]
  by: Closed-loop poles with unity negative feedback  (roots of den(s) + K num(s) by the
Aberth-Ehrlich iteration)
  assumes: a proper G(s); unity negative feedback
```

**What the worked example means:** This calculation determines the poles of a unity negative feedback control system by finding the roots of the characteristic polynomial formed by adding the open-loop denominator to the product of the gain and the open-loop numerator. The resulting pole locations, listed as [-2, -1], define the dynamic behavior of the system. The stability is assessed by examining the real parts of these poles; specifically, if the maximum real part -1 is negative, the system is considered stable. The characteristic polynomial itself is given by [1, 3, 2].

### Controllability and observability of a single-input single-output system, `controllability_observability`

The ranks of the controllability matrix [B AB ... A^(n-1)B] and the observability matrix [C; CA; ...; CA^(n-1)], and whether the system is controllable and observable.

$$\mathcal{C} = [B\ AB\ \cdots\ A^{n-1}B],\ \mathcal{O} = [C;\ CA;\ \cdots;\ CA^{n-1}]$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `A` | in | state matrix (n x n) | , |
| `B` | in | input vector | , |
| `C` | in | output vector | , |
| `controllability_rank` | out | rank of the controllability matrix | , |
| `observability_rank` | out | rank of the observability matrix | , |
| `controllable` | out | full controllability rank | , |
| `observable` | out | full observability rank | , |

**Applies when:** small integer or well-scaled entries (rank decided at a relative tolerance of 1e-9).

```
$ electrical-engineering solve controllability_observability "A=[[0, 1], [-2, -3]]" "B=[0,
1]" "C=[1, 0]"
controllability_rank = 2
observability_rank = 2
controllable = true
observable = true
  by: Controllability and observability of a single-input single-output system  (Kalman
rank tests by Gaussian elimination with partial pivoting)
  assumes: small integer or well-scaled entries (rank decided at a relative tolerance of
1e-9)
```

**What the worked example means:** This calculation determines the structural properties of a linear dynamic system by constructing the controllability and observability matrices from the state matrix A, input vector B, and output vector C. The rank of the controllability matrix, found to be 2, indicates how many independent

states can be influenced by the input. Similarly, the rank of the observability matrix, which is 2, shows how many independent states can be inferred from the output. Based on these ranks compared to the system dimension, the system is classified as yes and yes, which are essential conditions for designing full-state feedback controllers and state observers.

**Frequency response (Bode magnitude and phase) at one frequency, `bode_point`**

|G(jw)| as a ratio and in decibels, and the phase of G(jw), for a transfer function at an angular frequency w.

$$G(j\omega) = \frac{N(j\omega)}{D(j\omega)}, \quad 20\log_{10}|G(j\omega)|$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `num` | in | numerator coefficients of the open-loop G(s), highest power of s first | , |
| `den` | in | denominator coefficients of G(s), highest power of s first | , |
| `w` | in | angular frequency in rad/s | rad/s |
| `magnitude` | out | \|G(jw)\| | , |
| `magnitude_db` | out | 20 log10 \|G(jw)\| in decibels | , |
| `phase_deg` | out | phase of G(jw) in degrees, in (-180, 180] | , |
| `real` | out | real part of G(jw) | , |
| `imag` | out | imaginary part of G(jw) | , |

**Applies when:** no pole of G on the imaginary axis at w; w > 0.

```
$ electrical-engineering solve bode_point "num=[10]" "den=[1, 1]" "w=1 rad/s"
magnitude = 7.071067812
magnitude_db = 16.98970004
phase_deg = -45
real = 5
imag = -5
  by: Frequency response (Bode magnitude and phase) at one frequency  (Horner evaluation of
numerator and denominator at s = jw)
  w = 1 rad/s
  assumes: no pole of G on the imaginary axis at w; w > 0
```

**What the worked example means:** This calculation determines the frequency response of a system at the specific angular frequency 1 rad/s rad/s. Substituting this value into the transfer function defined by numerator coefficients [10] and denominator coefficients [1, 1] gives the complex gain. The resulting magnitude is 7.07107, which corresponds to 16.9897 on a decibel scale. The phase shift introduced by the system is -45 degrees, while the real and imaginary components are 5 and -5 respectively.

**Gain margin and phase-crossover frequency, `gain_margin`**

The gain margin of a unity-feedback loop with open-loop G(s): 1/|G(jw)| at the frequency where the phase is -180 degrees.

$$\angle G(j\omega_{pc}) = -180^\circ, \quad \mathrm{GM} = \frac{1}{|G(j\omega_{pc})|}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `num` | in | numerator coefficients of the open-loop G(s), highest power of s first | , |
| `den` | in | denominator coefficients of G(s), highest power of s first | , |
| `gain_margin` | out | gain margin as a ratio | , |
| `gain_margin_db` | out | gain margin in decibels | , |
| `phase_crossover` | out | phase-crossover frequency (phase -180 degrees) | rad/s |

**Applies when:** a proper G(s); exactly one phase crossover with G negative real.

```
$ electrical-engineering solve gain_margin "num=[10]" "den=[1, 6, 5, 0]"
gain_margin = 3
gain_margin_db = 9.542425094
phase_crossover = 2.236067977 rad/s
  by: Gain margin and phase-crossover frequency  (the phase crossover as the positive real
root of Im(N(jw) conj D(jw)) with Re G < 0 (Aberth iteration, Newton-polished))
  assumes: a proper G(s); exactly one phase crossover with G negative real
```

**What the worked example means:** This calculation determines the gain margin and phase crossover frequency for a unity feedback system with open-loop transfer function G(s). The phase crossover frequency, 2.23607 rad/s, is the positive real root where the imaginary part of the numerator times the conjugate of the denominator equals zero and the real part of G is negative. At this frequency, the phase angle is minus one hundred eighty degrees. The gain margin, expressed as 3 or 9.54243, represents the factor by which the open-loop gain can be increased before the closed-loop system becomes unstable.

### Maximum phase lead of a lead compensator, `lead_compensator`

For C(s) = (s + z)/(s + p) with p > z: the maximum phase lead, the frequency where it occurs, and the gain there.

$$\phi_{max} = \sin^{-1}\frac{p-z}{p+z},\ \omega_m = \sqrt{zp}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `z` | in | zero (corner) frequency | rad/s |
| `p` | in | pole (corner) frequency, p > z | rad/s |
| `phase_max_deg` | out | maximum phase lead in degrees | , |
| `w_max` | out | frequency of maximum phase, sqrt(z p) | rad/s |
| `gain_at_w_max_db` | out | gain of (s+z)/(s+p) there, in dB | , |

**Applies when:** p > z > 0.

```
$ electrical-engineering solve lead_compensator "z=1 rad/s" "p=10 rad/s"
phase_max_deg = 54.90319877
w_max = 3.16227766 rad/s
gain_at_w_max_db = -10
  by: Maximum phase lead of a lead compensator  (phi_max = asin((p - z)/(p + z)) at w =
sqrt(z p))
  z = 1 rad/s
  p = 10 rad/s
  assumes: p > z > 0
```

**What the worked example means:** This calculation determines the peak phase contribution of a lead compensator defined by a zero at 1 rad/s and a pole at 10 rad/s. The maximum phase lead occurs at the geometric mean frequency 3.16228 rad/s, yielding a phase shift of 54.9032. At this specific frequency, the magnitude of the transfer function is -10, which reflects the attenuation introduced by the network.

**Phase margin and gain-crossover frequency, `phase_margin`**

The phase margin of a unity-feedback loop with open-loop G(s): 180 degrees plus the phase of G at the frequency where |G(jw)| = 1.

$$|G(j\omega_c)| = 1, \quad \mathrm{PM} = 180^\circ + \angle G(j\omega_c)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `num` | in | numerator coefficients of the open-loop G(s), highest power of s first | , |
| `den` | in | denominator coefficients of G(s), highest power of s first | , |
| `phase_margin` | out | phase margin in degrees | , |
| `gain_crossover` | out | gain-crossover frequency (\|G\| = 1) | rad/s |

**Applies when:** a proper G(s); exactly one gain crossover.

```
$ electrical-engineering solve phase_margin "num=[4]" "den=[1, 2, 0]"
phase_margin = 51.82729237
gain_crossover = 1.572302756 rad/s
  by: Phase margin and gain-crossover frequency  (the gain crossover as the positive real
root of |N(jw)|^2 - |D(jw)|^2 (Aberth iteration, Newton-polished); PM = (arg G(jw_c) mod
360) - 180)
  assumes: a proper G(s); exactly one gain crossover
```

**What the worked example means:** This calculation determines the stability margins for a unity feedback system defined by the numerator coefficients [4] and denominator coefficients [1, 2, 0]. The gain crossover frequency, where the open-loop magnitude equals one, is found to be 1.5723 rad/s rad/s. At this specific frequency, the phase margin is computed as 51.8273 degrees, representing the amount of additional phase lag required to bring the system to the verge of instability.

**Routh-Hurwitz stability test, `routh_hurwitz`**

The Routh array's first column, the number of roots in the right half plane and whether a characteristic polynomial is stable.

$$b_i = \frac{a_1\, a_{0,i+1} - a_0\, a_{1,i+1}}{a_1}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `coefficients` | in | characteristic polynomial coefficients, highest power of s first | , |
| `rhp_roots` | out | number of roots with positive real part (sign changes in the first column) | , |
| `stable` | out | true when every root has a negative real part | , |
| `first_column` | out | the first column of the Routh array | , |

**Applies when:** no zero in the first column of the Routh array (the special cases are not handled).

```
$ electrical-engineering solve routh_hurwitz "coefficients=[1, 2, 3, 4, 5]"
rhp_roots = 2
stable = false
first_column = [1.0,2.0,1.0,-6.0,5.0]
  by: Routh-Hurwitz stability test  (Routh array built row by row; sign changes of its
first column counted)
  assumes: no zero in the first column of the Routh array (the special cases are not
handled)
```

**What the worked example means:** This calculation constructs the Routh array from the characteristic polynomial coefficients provided as input. Examining the signs of the elements in the first column gives the number of sign changes. The result shows that the first column contains values [1, 2, 1, -6, 5]. There are 2 sign changes, which indicates the number of poles with positive real parts. Consequently, the system is determined to be stable if no is true.

**Steady-state error of a unity-feedback system, `steady_state_error`**

The system type, the error constant (Kp, Kv or Ka) and the steady-state error for a unit step, ramp or parabola input.

$$e_{ss} = \lim_{s \to 0} \frac{s\, R(s)}{1 + G(s)}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `num` | in | numerator coefficients of the open-loop G(s), highest power of s first | , |
| `den` | in | denominator coefficients of G(s), highest power of s first | , |
| `input` | in | the test input: unit step 1/s, unit ramp 1/s^2 or unit parabola t^2/2 (1/s^3) | , |
| `steady_state_error` | out | steady-state error (or 'infinite') | , |
| `error_constant` | out | position (Kp), velocity (Kv) or acceleration (Ka) error constant | , |
| `system_type` | out | number of open-loop poles at the origin | , |

**Applies when:** a stable closed loop; unity negative feedback; unit-amplitude test input.

```
$ electrical-engineering solve steady_state_error "num=[10]" "den=[1, 3, 2]" input=step
steady_state_error = 0.1666666667
error_constant = 5
system_type = 0
  by: Steady-state error of a unity-feedback system  (system type from the poles at the
origin; error constant lim s^k G(s); e_ss = 1/(1+Kp) for a step, 1/Kv for a ramp, 1/Ka for
a parabola)
  input = step
  assumes: a stable closed loop; unity negative feedback; unit-amplitude test input
```

In the worked example, num = [10], den = [1, 3, 2], input = step go through the equation above and give steady_state_error = 0.1666666667, error_constant = 5, system_type = 0. It applies only when a stable closed loop; unity negative feedback; unit-amplitude test input: outside that it says what is wrong instead of returning a number.

**Steady-state speed and current of a permanent-magnet DC motor, `dc_motor`**

Speed, current, powers and efficiency of a DC motor with armature resistance R, torque constant Kt, back-emf constant Ke, viscous friction b and load torque T at supply voltage V.

$$\omega = \frac{K_t V - R T}{K_t K_e + R b}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | armature supply voltage | V |
| `R` | in | armature resistance | ohm |
| `Kt` | in | torque constant in N*m/A | , |
| `Ke` | in | back-emf constant in V*s/rad | , |
| `T` | in | load torque | N*m |
| `b` | in | viscous friction in N*m*s/rad | , |
| `speed` | out | steady-state speed | rad/s |
| `rpm` | out | speed in revolutions per minute | , |
| `current` | out | armature current | A |
| `input_power` | out | electrical input power | W |
| `output_power` | out | mechanical output power to the load | W |
| `efficiency` | out | output over input power | , |

**Applies when:** steady state; linear model (no saturation, constant parameters).

```
$ electrical-engineering solve dc_motor "V=12 V" "R=1 ohm" Kt=0.05 Ke=0.05 "T=0.02 N*m"
b=0.0001
speed = 223.0769231 rad/s
rpm = 2130.2277
current = 0.8461538462 A
input_power = 10.15384615 W
output_power = 4.461538462 W
efficiency = 0.4393939394
  by: Steady-state speed and current of a permanent-magnet DC motor  (solve V = R I + Ke w
and Kt I = b w + T)
  V = 12 V
  R = 1 ohm
  Kt = 0.05
  Ke = 0.05
  T = 0.02 N*m
  b = 0.0001
  assumes: steady state; linear model (no saturation, constant parameters)
```

**What the worked example means:** This calculation determines the operating point of a permanent-magnet DC motor by solving the coupled electrical and mechanical equations. Given the supply voltage, resistance, torque constant, back-emf constant, friction, and load torque, the model computes the resulting angular velocity in radians per second and revolutions per minute, alongside the drawn current. The final speed of 223.077 rad/s and current of 0.846154 A indicate how the motor balances electromagnetic torque against mechanical losses and external load at equilibrium.

**Step-response specifications of a standard second-order system, `second_order_step`**

Percent overshoot, peak time, 10-90% rise time and exact 2% settling time of wn^2/(s^2 + 2 zeta wn s + wn^2) for a unit step.

$$M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}, \quad t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `zeta` | in | damping ratio (0 < zeta < 1) | , |
| `wn` | in | undamped natural frequency | rad/s |
| `percent_overshoot` | out | maximum overshoot in percent | , |
| `overshoot` | out | maximum overshoot as a fraction of the final value | , |
| `peak_time` | out | time of the first peak | s |
| `rise_time` | out | 10% to 90% rise time | s |
| `t10` | out | time to reach 10% of the final value | s |
| `t90` | out | time to reach 90% of the final value | s |
| `settling_time` | out | exact 2% settling time (last time the response leaves the band) | s |
| `damped_frequency` | out | damped natural frequency wn sqrt(1 - zeta^2) | rad/s |

**Applies when:** 0 < zeta < 1 (underdamped); unit step from rest; unity DC gain.

```
$ electrical-engineering solve second_order_step zeta=0.5 "wn=4 rad/s"
percent_overshoot = 16.30335348
overshoot = 0.1630335348
peak_time = 0.9068996821 s
rise_time = 0.4093932368 s
t10 = 0.122057324 s
t90 = 0.5314505608 s
settling_time = 2.019087243 s
damped_frequency = 3.464101615 rad/s
  by: Step-response specifications of a standard second-order system  (closed-form
overshoot and peak time; rise and settling times by bisection on the exact response (the
last 2% crossing lies between the last peak above the band and the following zero of the
error))
  zeta = 0.5
  wn = 4 rad/s
  assumes: 0 < zeta < 1 (underdamped); unit step from rest; unity DC gain
```

**What the worked example means:** The system response to a unit step input is characterized by its transient performance metrics. With a damping ratio of 0.5 and an undamped natural frequency of 4 rad/s, the maximum overshoot reaches 0.163034 of the final value, corresponding to 16.3034 percent. The first peak occurs at 0.9069 s, while the rise time from ten to ninety percent is 0.409393 s. The exact two percent settling time is calculated as 2.01909 s, indicating when the response permanently stays within the error band.

**Transfer function of a single-input single-output state-space model, `state_space_to_tf`**

G(s) = C (sI - A)^-1 B + D as numerator and denominator coefficients for x' = Ax + Bu, y = Cx + Du.

$$G(s) = C(sI - A)^{-1}B + D$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `A` | in | state matrix (n x n) | , |
| `B` | in | input vector (n entries) | , |
| `C` | in | output vector (n entries) | , |
| `D` | in | feedthrough | , |
| `num` | out | numerator coefficients (highest power first, length n + 1) | , |
| `den` | out | denominator coefficients det(sI - A), monic | , |
| `order` | out | system order n | , |

**Applies when:** a square A with B and C of matching size; coefficients below 1e-12 of the largest are written as 0.

```
$ electrical-engineering solve state_space_to_tf "B=[0, 1]" "C=[1, 0]" "A=[[0, 1], [-2,
-3]]" D=0
num = [0.0,0.0,1.0]
den = [1.0,3.0,2.0]
order = 2
  by: Transfer function of a single-input single-output state-space model  (Faddeev-
LeVerrier recursion for det(sI - A) and adj(sI - A))
  D = 0
  assumes: a square A with B and C of matching size; coefficients below 1e-12 of the
largest are written as 0
```

**What the worked example means:** This calculation derives the single-input single-output transfer function from the state-space matrices A, B, C, and D. The resulting numerator coefficients are [0, 0, 1] and the denominator coefficients are [1, 3, 2], which define the poles of the system. This representation allows for frequency-domain analysis of the dynamics described by the state matrix A with dimension 2.

## D.c. machines (4)

### D.C. shunt generator: field and armature currents, generated emf and powers, `dc_shunt_generator`

The field current, armature current, generated emf, output power, power generated and copper losses of a shunt-wound d.c. generator delivering a load current at a terminal voltage.

$$I_a = I_L + I_f, \quad E = V + I_a R_a$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | terminal voltage | V |
| `IL` | in | load current | A |
| `Ra` | in | armature resistance | ohm |
| `Rf` | in | shunt field resistance | ohm |
| `If` | out | field current V/Rf | A |
| `Ia` | out | armature current IL + If | A |
| `E` | out | generated emf V + Ia Ra | V |
| `P_out` | out | power delivered to the load V IL | W |
| `P_generated` | out | electrical power generated E Ia | W |
| `copper_loss` | out | armature and field copper losses | W |

**Applies when:** steady state; brush drop neglected (or included in Ra).

```
$ electrical-engineering solve dc_shunt_generator "V=250 V" "IL=100 A" "Ra=0.1 ohm" "Rf=125
ohm"
If = 2 A
Ia = 102 A
E = 260.2 V
P_out = 25000 W
P_generated = 26540.4 W
copper_loss = 1540.4 W
  by: D.C. shunt generator: field and armature currents, generated emf and powers  (If =
V/Rf; Ia = IL + If; E = V + Ia Ra)
  V = 250 V
  IL = 100 A
  Ra = 0.1 ohm
  Rf = 125 ohm
  assumes: steady state; brush drop neglected (or included in Ra)
```

In the worked example, V = 250 V, IL = 100 A, Ra = 0.1 ohm, Rf = 125 ohm go through the equation above and give If = 2 A, Ia = 102 A, E = 260.2 V, P_out = 25000 W. It applies only when steady state; brush drop neglected (or included in Ra): outside that it says what is wrong instead of returning a number.

**D.C. shunt motor: back emf, torque, output and efficiency, `dc_shunt_motor`**

The field and armature currents, back emf, power converted, electromagnetic torque, output power and efficiency of a shunt-wound d.c. motor taking a current I from a supply V at a speed n.

$$E = V - I_a R_a, \quad T = \frac{E I_a}{2\pi n}, \quad \eta = \frac{E I_a - P_{loss}}{V I}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | supply voltage | V |
| `I` | in | current taken from the supply | A |
| `Ra` | in | armature resistance | ohm |
| `Rf` | in | shunt field resistance | ohm |
| `n` | in | speed in rev/s (rpm accepted in questions) | , |
| `losses` | in | iron, friction and windage losses (0 when not given) (optional) | W |
| `If` | out | field current V/Rf | A |
| `Ia` | out | armature current I - If | A |
| `E` | out | back emf V - Ia Ra | V |
| `P_converted` | out | power converted to mechanical form E Ia | W |
| `torque` | out | electromagnetic torque E Ia/(2 pi n) | N*m |
| `P_out` | out | output power (converted power minus the stated losses) | W |
| `efficiency` | out | efficiency P_out/(V I) | , |

**Applies when:** steady state; brush drop neglected (or included in Ra).

```
$ electrical-engineering solve dc_shunt_motor "I=30 A" "V=220 V" "Ra=0.4 ohm" "Rf=110 ohm"
n=16.66666667 "losses=500 W"
If = 2 A
Ia = 28 A
E = 208.8 V
P_converted = 5846.4 W
torque = 55.82900755 N*m
P_out = 5346.4 W
efficiency = 0.8100606061
  by: D.C. shunt motor: back emf, torque, output and efficiency  (If = V/Rf; Ia = I - If; E
= V - Ia Ra; T = E Ia/(2 pi n))
  I = 30 A
  V = 220 V
  Ra = 0.4 ohm
  Rf = 110 ohm
  n = 16.66666667
  losses = 500 W
  assumes: steady state; brush drop neglected (or included in Ra)
```

In the worked example, I = 30 A, V = 220 V, Ra = 0.4 ohm, Rf = 110 ohm, n = 16.66666667, losses = 500 W go through the equation above and give If = 2 A, Ia = 28 A, E = 208.8 V, P_converted = 5846.4 W. It applies only when steady state; brush drop neglected (or included in Ra): outside that it says what is wrong instead of returning a number.

**Generated emf of a d.c. machine (lap or wave winding), `dc_machine_emf`**

The emf generated in the armature of a d.c. machine (the back emf of a motor) from its poles, flux per pole, armature conductors, speed and winding.

$$E = \frac{p\,\Phi\,n\,Z}{a}, \quad a = p\ (\text{lap}),\ a = 2\ (\text{wave})$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `poles` | in | number of poles (even) | , |
| `flux` | in | useful flux per pole | Wb |
| `Z` | in | number of armature conductors | , |
| `n` | in | speed in rev/s (rpm accepted in questions) | , |
| `winding` | in | armature winding: lap or wave | , |
| `E` | out | generated emf | V |
| `paths` | out | parallel paths: the number of poles for lap, 2 for wave | , |
| `conductor_emf` | out | average emf of one conductor, poles x flux x n | V |

**Applies when:** an even number of poles; speed in rev/s; simplex windings.

```
$ electrical-engineering solve dc_machine_emf poles=4 "flux=0.025 Wb" Z=600 n=20
winding=lap
E = 300 V
paths = 4
conductor_emf = 2 V
  by: Generated emf of a d.c. machine (lap or wave winding)  (E = poles flux n Z / a with a
= poles (lap) or 2 (wave))
  poles = 4
  flux = 0.025 Wb
  Z = 600
  n = 20
  winding = lap
  assumes: an even number of poles; speed in rev/s; simplex windings
```

In the worked example, poles = 4, flux = 0.025 Wb, Z = 600, n = 20, winding = lap go through the equation above and give E = 300 V, paths = 4, conductor_emf = 2 V. It applies only when an even number of poles; speed in rev/s; simplex windings: outside that it says what is wrong instead of returning a number.

**New speed of a d.c. motor after a change of load or flux, `dc_motor_speed_change`**

The back emfs before and after, and the new speed, of a d.c. motor on a constant supply whose armature current changes from Ia1 to Ia2 and whose flux changes by a ratio (speed proportional to E/flux).

$$\frac{n_2}{n_1} = \frac{E_2}{E_1} \cdot \frac{\Phi_1}{\Phi_2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | supply voltage across the armature | V |
| `Ra` | in | armature resistance | ohm |
| `Ia1` | in | first armature current | A |
| `n1` | in | first speed in rev/s (rpm accepted in questions) | , |
| `Ia2` | in | new armature current | A |
| `flux_ratio` | in | new flux over the first flux (1 when unchanged) (optional) | , |
| `E1` | out | first back emf | V |
| `E2` | out | new back emf | V |
| `n2` | out | new speed in rev/s | , |
| `n2_rpm` | out | new speed in rpm | , |
| `speed_change` | out | fractional change of speed n2/n1 - 1 | , |

**Applies when:** a constant supply voltage across the armature; brush drop neglected.

```
$ electrical-engineering solve dc_motor_speed_change "V=220 V" "Ra=0.5 ohm" "Ia1=20 A"
n1=16.66666667 "Ia2=40 A"
E1 = 210 V
E2 = 200 V
n2 = 15.87301588
n2_rpm = 952.3809526
speed_change = -0.04761904762
  by: New speed of a d.c. motor after a change of load or flux  (E = V - Ia Ra; n2/n1 =
(E2/E1)(phi1/phi2))
  V = 220 V
  Ra = 0.5 ohm
  Ia1 = 20 A
  n1 = 16.66666667
  Ia2 = 40 A
  assumes: a constant supply voltage across the armature; brush drop neglected
```

In the worked example, V = 220 V, Ra = 0.5 ohm, Ia1 = 20 A, n1 = 16.66666667, Ia2 = 40 A go through the equation above and give E1 = 210 V, E2 = 200 V, n2 = 15.87301587, n2_rpm = 952.3809524. It applies only when a constant supply voltage across the armature; brush drop neglected: outside that it says what is wrong instead of returning a number.

## DC circuits (18)

### Current divider, `current_divider`

How a current I splits between two resistors in parallel.

$$I_1 = I\frac{R_2}{R_1 + R_2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `I` | in | total current into the parallel pair | A |
| `R1` | in | first branch resistor | ohm |
| `R2` | in | second branch resistor | ohm |
| `I1` | out | current through R1 | A |
| `I2` | out | current through R2 | A |

**Applies when:** R1, R2 > 0.

```
$ electrical-engineering solve current_divider "I=6 A" "R1=4 ohm" "R2=2 ohm"
I1 = 2 A
I2 = 4 A
  by: Current divider  (I1 = I R2/(R1 + R2))
  I = 6 A
  R1 = 4 ohm
  R2 = 2 ohm
  assumes: R1, R2 > 0
```

**What the worked example means:** This calculation determines how the total input current of 6 A divides between two parallel resistors. The branch containing the first resistor with a value of 4 ohm carries 2 A, and

the branch with the second resistor valued at 2 ohm carries 4 A. These results show that higher resistance in a specific branch leads to lower current flow through that path, consistent with the principles of parallel circuits.

**Delta-to-wye (Pi-to-T) resistor transformation, `delta_to_wye`**

The three wye (star) resistances equivalent to a delta (triangle) of resistances between terminals a, b, c.

$$R_a = \frac{R_{ab} R_{ca}}{R_{ab} + R_{bc} + R_{ca}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Rab` | in | delta resistance between terminals a and b | ohm |
| `Rbc` | in | delta resistance between terminals b and c | ohm |
| `Rca` | in | delta resistance between terminals c and a | ohm |
| `Ra` | out | wye arm at terminal a | ohm |
| `Rb` | out | wye arm at terminal b | ohm |
| `Rc` | out | wye arm at terminal c | ohm |

**Applies when:** every resistance positive.

```
$ electrical-engineering solve delta_to_wye "Rab=10 ohm" "Rbc=20 ohm" "Rca=30 ohm"
Ra = 5 ohm
Rb = 3.333333333 ohm
Rc = 10 ohm
  by: Delta-to-wye (Pi-to-T) resistor transformation  (each arm is the product of the two
adjacent delta resistances over their sum)
  Rab = 10 ohm
  Rbc = 20 ohm
  Rca = 30 ohm
  assumes: every resistance positive
```

**What the worked example means:** This calculation converts a delta network with resistances 10 ohm, 20 ohm, and 30 ohm into an equivalent star network. The resulting wye arms are 5 ohm at terminal a, 3.33333 ohm at terminal b, and 10 ohm at terminal c. Each wye resistance represents the equivalent value seen from its respective terminal to the common neutral point when viewed against the other two terminals.

**Equivalent capacitance of capacitors in parallel, `parallel_capacitance`**

The single capacitance equivalent to two or more capacitors connected in parallel.

$$C_{eq} = \sum_i C_i$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | the capacitance values, all in one unit (F, mF, uF, nF ...; a list, e.g. '10, 22, 47') | , |
| `unit` | in | the unit the values are written in (the result is in the same unit) (optional) | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `equivalent` | out | equivalent capacitance, in the unit of the inputs | , |
| `count` | out | number of capacitors | , |
| `unit` | out | the unit of the result | , |

**Applies when:** every value positive; at least two values.

```
$ electrical-engineering solve parallel_capacitance "C=[3, 6]" unit=uF
equivalent = 9
count = 2
unit = uF
  by: Equivalent capacitance of capacitors in parallel  (the sum of the values)
  unit = uF
  assumes: every value positive; at least two values
```

**What the worked example means:** This calculation determines the single equivalent capacitance for two or more capacitors connected in parallel. Adding the individual capacitance values together gives a total capacitance that represents the combined effect of all components sharing common terminals. The result, 9, is expressed in the same unit as the input values, ensuring consistency across the calculation.

### Equivalent capacitance of capacitors in series, `series_capacitance`

The single capacitance equivalent to two or more capacitors connected in series.

$$\frac{1}{C_{eq}} = \sum_{i} \frac{1}{C_i}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `C` | in | the capacitance values, all in one unit (F, mF, uF, nF ...; a list, e.g. '10, 22, 47') | , |
| `unit` | in | the unit the values are written in (the result is in the same unit) (optional) | , |
| `equivalent` | out | equivalent capacitance, in the unit of the inputs | , |
| `count` | out | number of capacitors | , |
| `unit` | out | the unit of the result | , |

**Applies when:** every value positive; at least two values.

```
$ electrical-engineering solve series_capacitance "C=[3, 6]" unit=uF
equivalent = 2
count = 2
unit = uF
  by: Equivalent capacitance of capacitors in series  (the reciprocal of the sum of
reciprocals)
  unit = uF
  assumes: every value positive; at least two values
```

**What the worked example means:** This calculation determines the single equivalent capacitance for capacitors connected in a series configuration. Taking the reciprocal of the sum of the reciprocals of each individual capacitance value shows that the total effective capacitance is 2. This result indicates that the

combined storage capacity is lower than that of any individual capacitor in the chain, which is essential for designing timing circuits and filters where reduced charge storage is required.

**Equivalent inductance of inductors in parallel, `parallel_inductance`**

The single inductance equivalent to two or more inductors connected in parallel.

$$\frac{1}{L_{eq}} = \sum_i \frac{1}{L_i}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L` | in | the inductance values, all in one unit (H, mH, uH, nH ...; a list, e.g. '10, 22, 47') | , |
| `unit` | in | the unit the values are written in (the result is in the same unit) (optional) | , |
| `equivalent` | out | equivalent inductance, in the unit of the inputs | , |
| `count` | out | number of inductors | , |
| `unit` | out | the unit of the result | , |

**Applies when:** every value positive; at least two values.

```
$ electrical-engineering solve parallel_inductance "L=[10, 40]" unit=mH
equivalent = 8
count = 2
unit = mH
  by: Equivalent inductance of inductors in parallel  (the reciprocal of the sum of
reciprocals)
  unit = mH
  assumes: every value positive; at least two values
```

In the worked example, L = [10, 40], unit = mH go through the equation above and give equivalent = 8, count = 2, unit = mH. It applies only when every value positive; at least two values: outside that it says what is wrong instead of returning a number.

**Equivalent inductance of inductors in series, `series_inductance`**

The single inductance equivalent to two or more inductors connected in series.

$$L_{eq} = \sum_i L_i$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L` | in | the inductance values, all in one unit (H, mH, uH, nH ...; a list, e.g. '10, 22, 47') | , |
| `unit` | in | the unit the values are written in (the result is in the same unit) (optional) | , |
| `equivalent` | out | equivalent inductance, in the unit of the inputs | , |
| `count` | out | number of inductors | , |
| `unit` | out | the unit of the result | , |

**Applies when:** every value positive; at least two values.

```
$ electrical-engineering solve series_inductance "L=[10, 22]" unit=mH
equivalent = 32
count = 2
unit = mH
  by: Equivalent inductance of inductors in series  (the sum of the values)
  unit = mH
  assumes: every value positive; at least two values
```

In the worked example, L = [10, 22], unit = mH go through the equation above and give equivalent = 32, count = 2, unit = mH. It applies only when every value positive; at least two values: outside that it says what is wrong instead of returning a number.

**Equivalent resistance of a series-parallel resistor network, `resistor_network`**

The equivalent resistance of a network written with + for series and || for parallel, e.g. 10 + (20 || 30).

$$R_s = R_1 + R_2, \quad R_p = \left( \frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `network` | in | the network in ohms: + joins in series, \|\| joins in parallel ('\|\|' binds tighter than '+'), parentheses group | , |
| `equivalent` | out | equivalent resistance | ohm |
| `resistors` | out | number of resistors in the network | , |

**Applies when:** every resistance positive; at least two resistors; '||' binds tighter than '+'.

```
$ electrical-engineering solve resistor_network "network=10 + (20 || 30)"
equivalent = 22 ohm
resistors = 3
  by: Equivalent resistance of a series-parallel resistor network  (recursive reduction of
the expression tree: series groups add, parallel groups add reciprocally)
  network = 10 + (20 || 30)
  assumes: every resistance positive; at least two resistors; '||' binds tighter than '+'
```

**What the worked example means:** This calculation determines the total resistance seen by a source connected to the resistor network defined as 10 + (20 || 30). By recursively combining series elements through addition and parallel elements through reciprocal addition, the complex arrangement reduces to a single value. The result, 22 ohm, represents the effective opposition to current flow for the entire circuit, while the count of individual components involved in this reduction is 3.

**Equivalent resistance of resistors in parallel, `parallel_resistance`**

The single resistance equivalent to two or more resistors connected in parallel.

$$\frac{1}{R_{eq}} = \sum_i \frac{1}{R_i}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | the resistance values in ohms (a list, e.g. '10, 20, 30') | , |
| `equivalent` | out | equivalent resistance | ohm |
| `count` | out | number of resistors | , |

**Applies when:** every value positive; at least two values.

```
$ electrical-engineering solve parallel_resistance "R=[10, 40]"
equivalent = 8 ohm
count = 2
  by: Equivalent resistance of resistors in parallel  (the reciprocal of the sum of
reciprocals)
  assumes: every value positive; at least two values
```

**What the worked example means:** This calculation determines the single resistance value that replaces multiple resistors connected in parallel. By applying the reciprocal of the sum of reciprocals method to the input values, the model computes an equivalent resistance of 8 ohm. This result represents the total opposition to current flow across the shared nodes, which is always lower than the smallest individual resistor in the group.

### Equivalent resistance of resistors in series, `series_resistance`

The single resistance equivalent to two or more resistors connected in series.

$$R_{eq} = \sum_{i} R_i$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | the resistance values in ohms (a list, e.g. '10, 20, 30') | , |
| `equivalent` | out | equivalent resistance | ohm |
| `count` | out | number of resistors | , |

**Applies when:** every value positive; at least two values.

```
$ electrical-engineering solve series_resistance "R=[10, 20, 30]"
equivalent = 60 ohm
count = 3
  by: Equivalent resistance of resistors in series  (the sum of the values)
  assumes: every value positive; at least two values
```

**What the worked example means:** This calculation determines the single resistance value that replaces a series of resistors. It sums the individual resistance values provided in [10, 20, 30] to produce the total equivalent resistance, reported as 60 ohm. This result represents the total opposition to current flow in a circuit where components are connected end-to-end.

### Maximum power transfer, `max_power_transfer`

The load resistance that draws the most power from a source with Thevenin voltage Vth and resistance Rth, and that power.

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Vth` | in | Thevenin (open-circuit) voltage | V |
| `Rth` | in | Thevenin resistance | ohm |
| `RL` | out | load resistance for maximum power (= Rth) | ohm |
| `Pmax` | out | maximum power delivered to the load | W |
| `efficiency` | out | fraction of the source power reaching the load at that point | , |

**Applies when:** a resistive load; Rth > 0.

```
$ electrical-engineering solve max_power_transfer "Vth=12 V" "Rth=6 ohm"
RL = 6 ohm
Pmax = 6 W
efficiency = 0.5
  by: Maximum power transfer  (RL = Rth, Pmax = Vth^2/(4 Rth))
  Vth = 12 V
  Rth = 6 ohm
  assumes: a resistive load; Rth > 0
```

**What the worked example means:** This calculation determines the optimal load resistance for extracting the greatest possible power from a source defined by its Thevenin voltage and resistance. By setting the load resistance equal to the Thevenin resistance, the system achieves maximum power transfer, resulting in a power output of 6 W at a load value of 6 ohm. At this specific operating point, the efficiency of power transfer is fixed at 0.5, meaning half of the total power generated is dissipated within the source itself rather than delivered to the load.

**Mesh currents from the mesh equations (Kirchhoff's voltage law), `mesh_currents`**

Solve the mesh equations R i = v of a circuit: the resistance matrix and the source-voltage vector give every mesh current.

$$\mathbf{R}\,\mathbf{i} = \mathbf{v}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `A` | in | mesh resistance matrix in ohms (self-resistances on the diagonal, minus the shared resistances off it) | , |
| `b` | in | net source voltage around each mesh in volts, in the mesh-current direction | , |
| `currents` | out | the mesh currents in amperes, in the order of the rows | , |
| `first` | out | the first mesh current | A |

**Applies when:** square matrix; one source voltage per mesh; a non-singular matrix.

```
$ electrical-engineering solve mesh_currents "A=[[10, -4], [-4, 8]]" "b=[12, 0]"
currents = [1.5,0.7500000000000001]
first = 1.5 A
  by: Mesh currents from the mesh equations (Kirchhoff's voltage law)  (Gaussian
elimination with partial pivoting)
  assumes: square matrix; one source voltage per mesh; a non-singular matrix
```

**What the worked example means:** This calculation determines the individual loop currents in a planar circuit by solving the system of linear equations formed from Kirchhoff's voltage law. The resistance matrix and source voltage vector define how electrical energy drives current through each independent path. The resulting values represent the magnitude and direction of flow for every mesh, expressed as [1.5, 0.75] amperes.

**Node voltages from the nodal equations (Kirchhoff's current law), `nodal_voltages`**

Solve the nodal equations G v = i of a circuit: the conductance matrix and the injected-current vector give every node voltage.

$$\mathbf{G}\,\mathbf{v} = \mathbf{i}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `A` | in | nodal conductance matrix in siemens (sums of conductances on the diagonal, minus the shared conductances off it) | , |
| `b` | in | current injected into each node by sources, in amperes | , |
| `voltages` | out | the node voltages in volts, in the order of the rows | , |
| `first` | out | the first node voltage | V |

**Applies when:** square matrix; one injected current per node; a non-singular matrix.

```
$ electrical-engineering solve nodal_voltages "A=[[0.5, -0.25], [-0.25, 0.75]]" "b=[2, 0]"
voltages = [4.8,1.6]
first = 4.8 V
  by: Node voltages from the nodal equations (Kirchhoff's current law)  (Gaussian
elimination with partial pivoting)
  assumes: square matrix; one injected current per node; a non-singular matrix
```

**What the worked example means:** This calculation determines the electrical potential at each node in a circuit by solving a system of linear equations based on Kirchhoff's Current Law. The inputs consist of the nodal conductance matrix, which represents how easily current flows between nodes, and the vector of injected currents from sources. By applying Gaussian elimination with partial pivoting to these values, the method computes the steady-state voltage at every node. The resulting voltages, such as 4.8 V for the first node, indicate the electric potential relative to the reference ground, allowing for the analysis of current flow through individual branches.

**Node voltages of a DC resistive circuit from its netlist, `dc_node_voltages`**

Every node voltage, every voltage-source current and the power delivered, for a circuit of resistors and independent sources written as a netlist.

$$\begin{bmatrix} G & B \\ B^T & 0 \end{bmatrix} \begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} i_s \\ e \end{bmatrix}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `netlist` | in | elements separated by ';': R<name> <node> <node> <ohms>, V<name> <node+> <node-> <volts>, I<name> <node> <node> <amperes> (flowing from the first node through the source to the second); node 0 is ground | , |
| `node_voltages` | out | voltage of every node relative to node 0, in volts | , |
| `source_currents` | out | current through each voltage source from its + node to its - node, in amperes | , |
| `power_delivered` | out | total power delivered by the sources (= power dissipated in the resistors) | W |

**Applies when:** a ground node 0; every resistance positive; a unique solution (no floating node, no loop of voltage sources).

```
$ electrical-engineering solve dc_node_voltages "netlist=V1 1 0 12; R1 1 2 1000; R2 2 0
2000."
node_voltages = {"1":12.0,"2":8.0}
source_currents = {"V1":-0.004}
power_delivered = 0.048 W
  by: Node voltages of a DC resistive circuit from its netlist  (modified nodal analysis
solved by Gaussian elimination)
  netlist = V1 1 0 12; R1 1 2 1000; R2 2 0 2000.
  assumes: a ground node 0; every resistance positive; a unique solution (no floating node,
no loop of voltage sources)
```

**What the worked example means:** This calculation determines the electrical state of a DC resistive circuit defined by its netlist. By applying modified nodal analysis, it solves for the voltage at every node relative to ground, represented as {1: 12, 2: 8}. It also computes the current flowing through each independent voltage source, given by {V1: -0.004}, and sums the energy supplied by all sources, which equals the total dissipation in the resistors, shown as 0.048 W. These values provide a complete description of how the circuit distributes potential and current under steady-state conditions.

### Norton equivalent of a DC circuit between two nodes, `norton_equivalent`

The Norton current and resistance seen between two nodes of a resistive circuit with independent sources.

$$I_N = \frac{V_{th}}{R_{th}}, \quad R_N = R_{th}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `netlist` | in | elements separated by ';': R<name> <node> <node> <ohms>, V<name> <node+> <node-> <volts>, I<name> <node> <node> <amperes>; node 0 is ground | , |
| `node_a` | in | the terminal node (the + side of the equivalent) | , |
| `node_b` | in | the reference terminal node | , |
| `In` | out | Norton (short-circuit) current from node a to node b | A |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Rn` | out | Norton resistance (equal to the Thevenin resistance) | ohm |

**Applies when:** both nodes in the circuit; no ideal voltage source directly across the terminals; a unique solution (no floating node).

```
$ electrical-engineering solve norton_equivalent node_a=2 node_b=0 "netlist=V1 1 0 12; R1 1
2 1000; R2 2 0 2000."
In = 0.012 A
Rn = 666.6666667 ohm
  by: Norton equivalent of a DC circuit between two nodes  (Thevenin voltage and resistance
by modified nodal analysis, then I_N = V_th / R_th)
  netlist = V1 1 0 12; R1 1 2 1000; R2 2 0 2000.
  node_a = 2
  node_b = 0
  assumes: both nodes in the circuit; no ideal voltage source directly across the
terminals; a unique solution (no floating node)
```

In the worked example, node_a = 2, node_b = 0, netlist = V1 1 0 12; R1 1 2 1000; R2 2 0 2000. go through the equation above and give In = 0.012 A, Rn = 666.6666667 ohm. It applies only when both nodes in the circuit; no ideal voltage source directly across the terminals; a unique solution (no floating node): outside that it says what is wrong instead of returning a number.

**Thevenin equivalent of a DC circuit between two nodes, `thevenin_equivalent`**

The Thevenin voltage and resistance seen between two nodes of a resistive circuit with independent sources.

$$V_{th} = v_a - v_b\big|_{open}, \quad R_{th} = \frac{v_a - v_b}{1\,\mathrm{A}}\Big|_{sources\ off}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `netlist` | in | elements separated by ';': R<name> <node> <node> <ohms>, V<name> <node+> <node-> <volts>, I<name> <node> <node> <amperes>; node 0 is ground | , |
| `node_a` | in | the terminal node (the + side of the equivalent) | , |
| `node_b` | in | the reference terminal node | , |
| `Vth` | out | Thevenin (open-circuit) voltage, node a relative to node b | V |
| `Rth` | out | Thevenin resistance | ohm |

**Applies when:** both nodes in the circuit; no ideal voltage source directly across the terminals; a unique solution (no floating node).

```
$ electrical-engineering solve thevenin_equivalent node_a=2 node_b=0 "netlist=V1 1 0 12; R1
1 2 1000; R2 2 0 2000."
Vth = 8 V
Rth = 666.6666667 ohm
  by: Thevenin equivalent of a DC circuit between two nodes  (open-circuit voltage by
modified nodal analysis; resistance by a 1 A test current with the independent sources set
to zero)
  netlist = V1 1 0 12; R1 1 2 1000; R2 2 0 2000.
  node_a = 2
  node_b = 0
  assumes: both nodes in the circuit; no ideal voltage source directly across the
terminals; a unique solution (no floating node)
```

**What the worked example means:** This calculation finds the Thevenin equivalent circuit seen from two specific terminals. It determines the open-circuit voltage between node a and node b, denoted as 8 V, and the equivalent resistance, 666.667 ohm, by deactivating independent sources and applying a test current. The result represents a simplified linear model of the complex network, allowing easy analysis of how any load connected to these terminals will behave.

**Voltage divider, `voltage_divider`**

The output voltage across R2 of two resistors in series across Vin (R1 on top, R2 to ground), and the current.

$$V_{out} = V_{in}\frac{R_2}{R_1 + R_2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Vin` | in | input voltage | V |
| `R1` | in | top resistor (from the input to the output) | ohm |
| `R2` | in | bottom resistor (from the output to ground) | ohm |
| `Vout` | out | output voltage across R2 (unloaded) | V |
| `I` | out | current through the divider | A |

**Applies when:** no load on the output; R1, R2 > 0.

```
$ electrical-engineering solve voltage_divider "Vin=12 V" "R1=1000 ohm" "R2=2000 ohm"
Vout = 8 V
I = 0.004 A
  by: Voltage divider  (Vout = Vin R2 / (R1 + R2))
  Vin = 12 V
  R1 = 1000 ohm
  R2 = 2000 ohm
  assumes: no load on the output; R1, R2 > 0
```

**What the worked example means:** This calculation determines the voltage at the node between two series resistors connected across an input source. Using the input voltage of 12 V and resistor values of 1000 ohm for the top element and 2000 ohm for the bottom element, the resulting output voltage is 8 V. The total current flowing through this path is 0.004 A. This result assumes no external load is connected to the output node, meaning the measured voltage represents the ideal potential difference across the lower resistor.

### Wheatstone bridge output, `wheatstone_bridge`

The output voltage of a Wheatstone bridge (left arm R1 over R2, right arm R3 over R4, supply Vs), whether it is balanced, and the R4 that balances it.

$$V_{out} = V_s \left( \frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Vs` | in | bridge supply voltage | V |
| `R1` | in | left arm, top | ohm |
| `R2` | in | left arm, bottom | ohm |
| `R3` | in | right arm, top | ohm |
| `R4` | in | right arm, bottom | ohm |
| `Vout` | out | voltage of the left midpoint relative to the right midpoint | V |
| `balanced` | out | true when R1 R4 = R2 R3 | , |
| `R4_balance` | out | the R4 that would balance the bridge | ohm |

**Applies when:** an unloaded output (ideal voltmeter).

```
$ electrical-engineering solve wheatstone_bridge "Vs=10 V" "R1=100 ohm" "R2=200 ohm"
"R3=150 ohm" "R4=250 ohm"
Vout = 0.4166666667 V
balanced = false
R4_balance = 300 ohm
  by: Wheatstone bridge output  (difference of the two divider outputs)
  Vs = 10 V
  R1 = 100 ohm
  R2 = 200 ohm
  R3 = 150 ohm
  R4 = 250 ohm
  assumes: an unloaded output (ideal voltmeter)
```

**What the worked example means:** This calculation determines the voltage difference between the midpoints of a Wheatstone bridge using the supply voltage and four resistor values. The result, 0.416667 V, represents the potential at the left midpoint relative to the right midpoint. If this value is zero, the bridge is balanced, meaning the ratio of resistors in the left arm matches the ratio in the right arm. When unbalanced, the magnitude of 0.416667 V indicates the degree of deviation from equilibrium.

### Wye-to-delta (T-to-Pi) resistor transformation, `wye_to_delta`

The three delta (triangle) resistances equivalent to a wye (star) of resistances at terminals a, b, c.

$$R_{ab} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_c}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Ra` | in | wye arm at terminal a | ohm |
| `Rb` | in | wye arm at terminal b | ohm |
| `Rc` | in | wye arm at terminal c | ohm |
| `Rab` | out | delta resistance between a and b | ohm |
| `Rbc` | out | delta resistance between b and c | ohm |
| `Rca` | out | delta resistance between c and a | ohm |

**Applies when:** every resistance positive.

```
$ electrical-engineering solve wye_to_delta "Ra=5 ohm" "Rb=10 ohm" "Rc=20 ohm"
Rab = 17.5 ohm
Rbc = 70 ohm
Rca = 35 ohm
  by: Wye-to-delta (T-to-Pi) resistor transformation  (each delta resistance is the sum of
pairwise products of the arms over the opposite arm)
  Ra = 5 ohm
  Rb = 10 ohm
  Rc = 20 ohm
  assumes: every resistance positive
```

**What the worked example means:** This calculation determines the equivalent delta resistances for a given star configuration. Using the wye arm values 5 ohm, 10 ohm, and 20 ohm, the method computes each line-to-line resistance by summing the pairwise products of the arms and dividing by the opposite arm. For instance, the resistance between terminals a and b results in 17.5 ohm, representing the direct connection in the triangular topology that behaves identically to the original star arrangement at these three nodes.

## Digital logic (9)

### Analog-to-digital conversion: output code and resolution, `adc_conversion`

The output code (truncating converter), its binary form, the LSB size and the quantized voltage of an ideal n-bit ADC with reference Vref.

$$\text{code} = \left\lfloor \frac{V_{in}}{V_{ref}/2^{n}} \right\rfloor$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Vin` | in | input voltage (0 to Vref) | V |
| `Vref` | in | full-scale reference voltage | V |
| `bits` | in | resolution in bits | , |
| `code` | out | output code | , |
| `binary` | out | code in binary | , |
| `lsb` | out | LSB size Vref/2^n | V |
| `lsb_mV` | out | LSB in millivolts | , |
| `quantized` | out | voltage of the code, code x LSB | V |

**Applies when:** 0 <= Vin <= Vref; truncating (floor) quantizer; Vref itself gives the top code.

```
$ electrical-engineering solve adc_conversion "Vin=1.2 V" "Vref=5 V" bits=8
code = 61
binary = 00111101
lsb = 0.01953125 V
lsb_mV = 19.53125
quantized = 1.19140625 V
  by: Analog-to-digital conversion: output code and resolution  (code = floor(Vin / LSB),
capped at 2^n - 1)
  Vin = 1.2 V
  Vref = 5 V
  assumes: 0 <= Vin <= Vref; truncating (floor) quantizer; Vref itself gives the top code
```

**What the worked example means:** This calculation determines the digital representation of an analog input voltage using an ideal n-bit truncating converter. Given an input voltage of 1.2 V and a reference voltage of 5 V, the least significant bit size is calculated as 0.0195312 V. The resulting output code is 61, which corresponds to a quantized voltage of 1.19141 V. This process maps the continuous input signal to discrete digital levels, illustrating the resolution limits inherent in analog-to-digital conversion.

### Binary to Gray code and Gray code to binary, `gray_code`

The reflected binary (Gray) code of a binary word, or the binary word of a Gray code, with the same width.

$$g = b \oplus (b \gg 1)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `value` | in | the bits | , |
| `direction` | in | conversion direction | , |
| `result` | out | the converted bits | , |
| `decimal` | out | the result read as an unsigned number | , |

**Applies when:** 1 to 32 bits.

```
$ electrical-engineering solve gray_code value=1011 direction=to_gray
result = 1110
decimal = 14
  by: Binary to Gray code and Gray code to binary  (g = b XOR (b >> 1); back: b = g XOR (g
>> 1) XOR (g >> 2) ...)
  value = 1011
  direction = to_gray
  assumes: 1 to 32 bits
```

**What the worked example means:** This calculation converts the provided bit pattern 1011 into its corresponding reflected binary Gray code 1110. The conversion applies an exclusive OR operation between the original binary word and a version shifted right by one position. This method ensures that the resulting sequence maintains the property where consecutive numbers differ by exactly one bit, which is essential for minimizing errors in state transitions. The output 14 represents this converted bit pattern interpreted as an unsigned integer value.

### Equivalence of two Boolean expressions, `boolean_equivalence`

Whether two Boolean expressions are equal for every input, with a counterexample row when they are not.

$$F_1 \equiv F_2 \iff F_1 \oplus F_2 = 0$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `expression1` | in | first expression (same notation as boolean_evaluate) | , |
| `expression2` | in | second expression | , |
| `equivalent` | out | true when the two agree on every row | , |
| `counterexample` | out | the first row where they differ ('none' if equivalent) | , |

**Applies when:** at most 12 variables.

```
$ electrical-engineering solve boolean_equivalence "expression1=(A·B)'" "expression2=A' + B'"
equivalent = true
counterexample = none
  by: Equivalence of two Boolean expressions  (exhaustive comparison over the truth table)
  expression1 = (A·B)'
  expression2 = A' + B'
  assumes: at most 12 variables
```

**What the worked example means:** This calculation checks whether the two provided Boolean expressions, (A·B)' and A' + B', yield identical results for every possible combination of their variables. The method involves an exhaustive comparison over the truth table, verifying that the exclusive OR of the two functions equals zero. The final determination is recorded as yes, which indicates if the expressions are logically equal. If they are not equivalent, the system identifies a specific input row where they differ, reported as none. This process ensures that circuit simplifications or design changes preserve the original logical behavior.

### Evaluate a Boolean expression, `boolean_evaluate`

The value (0 or 1) of a Boolean expression for given values of its variables.

$$F(A, B, \ldots) \in \{0, 1\}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `expression` | in | the expression: AND (&, ·, *), OR (\|, +), XOR (^), NOT (~, !, NOT, or a postfix '), parentheses; NOT > AND > XOR > OR | , |
| `values` | in | the variables' values, e.g. 'A=1, B=0, C=1' | , |
| `value` | out | value of the expression (0 or 1) | , |

**Applies when:** every variable given 0 or 1; products written with an operator (A·B, not AB).

```
$ electrical-engineering solve boolean_evaluate "expression=A AND (B OR NOT C)"
"values=A=1, B=0, C=0"
value = 1
  by: Evaluate a Boolean expression  (recursive-descent parse with NOT > AND > XOR > OR,
then evaluation)
  expression = A AND (B OR NOT C)
  values = A=1, B=0, C=0
  assumes: every variable given 0 or 1; products written with an operator (A·B, not AB)
```

**What the worked example means:** This calculation determines the final logical state of a Boolean expression by applying standard precedence rules, where NOT operations are processed before AND, which precedes XOR, and finally OR. The process begins with the input expression A AND (B OR NOT C) and the specific binary values assigned to its variables, provided as A=1, B=0, C=0. Each variable is substituted with its given value, and the expression is parsed recursively to resolve parentheses and operators in the correct order. The resulting single bit, either zero or one, is reported as 1, representing the definitive truth value of the entire logic circuit or statement under those specific conditions.

**Minimal sum of products from minterms (with don't-cares), `minterms_to_sop`**

A minimal sum of products (fewest terms, then fewest literals) of F(A, B, C, D) = Σm(...) + d(...), the Karnaugh-map result.

$$F = \sum m(\ldots) + \sum d(\ldots)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `minterms` | in | minterm numbers (A is the most significant bit) | , |
| `variables` | in | number of variables (1 to 4) | , |
| `dont_cares` | in | don't-care rows (optional) | , |
| `minimal_sop` | out | a minimal sum of products in A, B, C, D (A' = NOT A) | , |
| `terms` | out | number of product terms | , |
| `literals` | out | number of literals | , |
| `variables` | out | the variables | , |

**Applies when:** 1 to 4 variables; minterms and don't-cares disjoint and in range.

```
$ electrical-engineering solve minterms_to_sop "minterms=[0, 1, 2, 5, 6, 7]" variables=3
minimal_sop = A'·B' + A·C + B·C'
terms = 3
literals = 6
variables = A, B, C
  by: Minimal sum of products from minterms (with don't-cares)  (Quine-McCluskey prime
implicants over minterms and don't-cares; exact minimum cover of the minterms)
  assumes: 1 to 4 variables; minterms and don't-cares disjoint and in range
```

In the worked example, minterms = [0, 1, 2, 5, 6, 7], variables = 3 go through the equation above and give minimal_sop = A'·B' + A·C + B·C', terms = 3, literals = 6, variables = A, B, C. It applies only when 1 to 4 variables; minterms and don't-cares disjoint and in range: outside that it says what is wrong instead of returning a number.

### Number base conversion (binary, octal, decimal, hexadecimal), `base_conversion`

A whole number written in base 2, 8, 10 or 16, given in all four bases.

$$N = \sum_i d_i \, b^i$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `value` | in | the digits (0b, 0o, 0x prefixes allowed) | , |
| `base` | in | the base the value is written in | , |
| `decimal` | out | value in base 10 | , |
| `binary` | out | base 2 | , |
| `octal` | out | base 8 | , |
| `hexadecimal` | out | base 16 (upper-case digits) | , |

**Applies when:** a non-negative whole number below 2^53; digits valid for the base.

```
$ electrical-engineering solve base_conversion value=1011 base=2
decimal = 11
binary = 1011
octal = 13
hexadecimal = B
  by: Number base conversion (binary, octal, decimal, hexadecimal)  (positional value, then
repeated division by the target base)
  value = 1011
  base = 2
  assumes: a non-negative whole number below 2^53; digits valid for the base
```

**What the worked example means:** This calculation determines the equivalent representation of a non-negative integer across four common numerical bases. Starting with the input value 1011 defined in base 2, the algorithm first computes its positional decimal value using the sum of digits multiplied by powers of the base. This decimal result 11 is then converted into binary, octal, and hexadecimal formats through repeated division by the target bases. The final outputs provide the same magnitude expressed as 1011, 13, and B, allowing engineers to view data at different levels of hardware abstraction.

### Truth table and minimal sum of products of a Boolean expression, `truth_table_sop`

The minterms (truth table), the canonical sum of products and a minimal sum of products (fewest terms, then fewest literals) of an expression in up to 4 variables.

$$F = \sum m(\dots)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `expression` | in | the expression: AND (&, ·, *), OR (\|, +), XOR (^), NOT (~, !, NOT, or a postfix '), parentheses; NOT > AND > XOR > OR | , |
| `variables` | out | the variables in truth-table order (first = most significant bit) | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `minterms` | out | rows where the expression is 1 | , |
| `maxterms` | out | rows where it is 0 | , |
| `canonical_sop` | out | sum of all minterms | , |
| `minimal_sop` | out | a minimal sum of products (A' means NOT A, · means AND) | , |
| `terms` | out | number of product terms in the minimal form | , |
| `literals` | out | number of literals in the minimal form | , |

**Applies when:** 1 to 4 variables.

```
$ electrical-engineering solve truth_table_sop "expression=A·B + A·B' + A'·B"
variables = A, B
minterms = [1.0,2.0,3.0]
maxterms = [0.0]
canonical_sop = A'·B + A·B' + A·B
minimal_sop = A + B
terms = 2
literals = 2
  by: Truth table and minimal sum of products of a Boolean expression  (truth table; Quine-McCluskey prime implicants; essential primes, then an exact search for the smallest cover)
  expression = A·B + A·B' + A'·B
  assumes: 1 to 4 variables
```

**What the worked example means:** This calculation determines the simplest logical form for the input expression A·B + A·B' + A'·B. It first generates a truth table to identify all rows where the output is one, listed as [1, 2, 3]. Using these values, it constructs the canonical sum of products and then applies the Quine-McCluskey algorithm to find prime implicants. The result is a minimized expression A + B that uses the fewest possible product terms 2 and literals 2, providing an efficient implementation for digital logic design.

### Two's complement representation of a signed integer, `twos_complement_encode`

The n-bit two's complement bit pattern (and its hexadecimal form) of a signed whole number.

$$x_{2c} = x \bmod 2^n$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `value` | in | the signed whole number | , |
| `bits` | in | word width in bits | , |
| `pattern` | out | the bit pattern, most significant bit first | , |
| `hexadecimal` | out | the pattern in hexadecimal | , |
| `unsigned` | out | the pattern read as an unsigned number | , |

**Applies when:** -2^(n-1) <= value < 2^(n-1).

```
$ electrical-engineering solve twos_complement_encode value=-5 bits=8
pattern = 11111011
hexadecimal = FB
unsigned = 251
  by: Two's complement representation of a signed integer  (value modulo 2^n (for a
negative number: invert the bits of |value| and add 1))
  assumes: -2^(n-1) <= value < 2^(n-1)
```

**What the worked example means:** This calculation determines the fixed-width binary representation of a signed integer using the two's complement system. Given the input value -5 and a word width of 8 bits, the resulting bit pattern is 11111011. When this specific sequence of bits is interpreted as an unsigned number, it yields the value 251. This method ensures that positive and negative integers share the same arithmetic rules at the hardware level, allowing subtraction to be performed via addition of the negated operand.

#### Value of a two's complement bit pattern, `twos_complement_decode`

The signed value of an n-bit two's complement pattern (n = the number of digits written).

$$x = -b_{n-1}2^{n-1} + \sum_{i<n-1} b_i 2^i$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `pattern` | in | the bits, most significant first | , |
| `value` | out | signed value | , |
| `bits` | out | word width | , |

**Applies when:** only 0 and 1; at most 64 bits.

```
$ electrical-engineering solve twos_complement_decode pattern=11111011
value = -5
bits = 8
  by: Value of a two's complement bit pattern  (unsigned value minus 2^n when the sign bit
is 1)
  pattern = 11111011
  assumes: only 0 and 1; at most 64 bits
```

**What the worked example means:** This calculation determines the signed integer represented by the binary sequence 11111011 using two's complement notation. The method interprets the most significant bit as the sign indicator; if it is one, the unsigned magnitude is reduced by twice the weight of that bit position to yield the negative result. The final output, -5, reflects this signed interpretation over a word width of 8 bits.

### Diodes and transistors (4)

#### DC operating point of a voltage-divider biased BJT, `bjt_divider_bias`

Base, collector and emitter currents and the node voltages of an NPN transistor with divider bias (R1 to VCC, R2 to ground), collector resistor RC and emitter resistor RE.

$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1) R_E}, \quad V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `VCC` | in | supply voltage | V |
| `R1` | in | divider resistor from VCC to the base | ohm |
| `R2` | in | divider resistor from the base to ground | ohm |
| `RC` | in | collector resistor | ohm |
| `RE` | in | emitter resistor | ohm |
| `beta` | in | DC current gain | , |
| `VBE` | in | base-emitter voltage in conduction (e.g. 0.7 V) | V |
| `IC` | out | collector current | A |
| `IC_mA` | out | collector current in milliamperes | , |
| `IB` | out | base current | A |
| `IE` | out | emitter current | A |
| `VB` | out | base voltage | V |
| `VE` | out | emitter voltage | V |
| `VC` | out | collector voltage | V |
| `VCE` | out | collector-emitter voltage | V |

**Applies when:** NPN in the forward-active region (VCE >= 0.2 V); constant VBE and beta.

```
$ electrical-engineering solve bjt_divider_bias "VCC=12 V" "VBE=0.7 V" "R1=10000 ohm"
"R2=2200 ohm" "RC=3600 ohm" "RE=1000 ohm" beta=100
IC = 0.001424015309 A
IC_mA = 1.424015309
IB = 1.424015309e-05 A
IE = 0.001438255462 A
VB = 2.138255462 V
VE = 1.438255462 V
VC = 6.873544889 V
VCE = 5.435289428 V
  by: DC operating point of a voltage-divider biased BJT  (Thevenin equivalent of the
divider, then IB = (VTH - VBE)/(RTH + (beta+1) RE) (the exact analysis, base current
included))
  VCC = 12 V
  VBE = 0.7 V
  R1 = 10000 ohm
  R2 = 2200 ohm
  RC = 3600 ohm
  RE = 1000 ohm
  beta = 100
  assumes: NPN in the forward-active region (VCE >= 0.2 V); constant VBE and beta
```

**What the worked example means:** This analysis determines the steady-state voltages and currents for an NPN transistor using voltage-divider bias. Finding the Thevenin equivalent of the resistor network gives the base current as 1.42402e-05 A, which sets the collector current to 0.00142402 A and the emitter current to

0.00143826 A. The resulting node voltages are base at 2.13826 V, emitter at 1.43826 V, and collector at 6.87354 V, confirming the device operates in the forward-active region with a collector-emitter voltage of 5.43529 V.

**MOSFET drain current in saturation (square law), `mosfet_saturation`**

Drain current, overdrive voltage and transconductance of an n-channel MOSFET in saturation.

$$I_D = \tfrac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_{TH})^2$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `kn` | in | process transconductance parameter mu_n Cox in A/V^2 | , |
| `WL` | in | aspect ratio W/L | , |
| `VGS` | in | gate-source voltage | V |
| `VTH` | in | threshold voltage | V |
| `VDS` | in | drain-source voltage | V |
| `ID` | out | drain current | A |
| `ID_mA` | out | drain current in milliamperes | , |
| `VOV` | out | overdrive VGS - VTH | V |
| `gm` | out | transconductance in A/V (siemens) | , |

**Applies when:** saturation: VGS > VTH and VDS >= VGS - VTH; no channel-length modulation.

```
$ electrical-engineering solve mosfet_saturation kn=0.0002 WL=10 "VGS=2 V" "VTH=0.7 V"
"VDS=5 V"
ID = 0.00169 A
ID_mA = 1.69
VOV = 1.3 V
gm = 0.0026
  by: MOSFET drain current in saturation (square law)  (ID = (1/2) kn (W/L) (VGS - VTH)^2)
  kn = 0.0002
  WL = 10
  VGS = 2 V
  VTH = 0.7 V
  VDS = 5 V
  assumes: saturation: VGS > VTH and VDS >= VGS - VTH; no channel-length modulation
```

**What the worked example means:** This calculation determines the operating point of an n-channel MOSFET in saturation. Using the process parameter 0.0002 and aspect ratio 10, along with gate-source voltage 2 V and threshold voltage 0.7 V, the overdrive voltage is found to be 1.3 V. Consequently, the drain current is 0.00169 A (1.69), and the small-signal transconductance is 0.0026. These values indicate how strongly the transistor conducts and responds to input changes under these bias conditions.

**Operating point of a diode in series with a resistor (Shockley equation), `diode_circuit`**

The current and the diode voltage of a forward-biased diode I = Is (exp(VD/(n VT)) - 1) in series with a resistor across a DC source.

$$V_s = IR + nV_T \ln\left(1 + \frac{I}{I_s}\right)$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Vs` | in | source voltage (forward) | V |
| `R` | in | series resistor | ohm |
| `Is` | in | saturation current | A |
| `n` | in | ideality factor | , |
| `VT` | in | thermal voltage kT/q (about 25.85 mV at 300 K) | V |
| `I` | out | circuit current | A |
| `I_mA` | out | current in milliamperes | , |
| `VD` | out | diode voltage | V |
| `VR` | out | resistor voltage | V |

**Applies when:** forward bias (Vs > 0); the Shockley model without series resistance or breakdown.

```
$ electrical-engineering solve diode_circuit "Vs=5 V" "R=1000 ohm" "Is=1e-14 A" n=1
"VT=0.02585 V"
I = 0.004307509625 A
I_mA = 4.307509625
VD = 0.6924903752 V
VR = 4.307509625 V
  by: Operating point of a diode in series with a resistor (Shockley equation)  (bisection
then Newton on Vs = I R + n VT ln(1 + I/Is))
  Vs = 5 V
  R = 1000 ohm
  Is = 1e-14 A
  n = 1
  VT = 0.02585 V
  assumes: forward bias (Vs > 0); the Shockley model without series resistance or breakdown
```

**What the worked example means:** This calculation determines the steady-state operating point for a forward-biased diode in series with a resistor. By solving the nonlinear circuit equation, it finds that the current through the branch is 4.30751 when driven by a source voltage of 5 V. The resulting voltage across the diode is 0.69249 V, while the remaining potential difference of 4.30751 V appears across the series resistance. This result reflects the balance between the exponential current-voltage characteristic of the semiconductor junction and the linear ohmic behavior of the resistor.

### Zener diode shunt regulator, `zener_regulator`

Load voltage and the currents of a Zener shunt regulator (series resistor R from the input, Zener and load in parallel).

$$I_Z = \frac{V_{in} - V_Z}{R} - \frac{V_Z}{R_L}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Vin` | in | input voltage | V |
| `R` | in | series resistor | ohm |
| `Vz` | in | Zener (breakdown) voltage | V |
| `RL` | in | load resistance | ohm |
| `VL` | out | load voltage | V |
| `IR` | out | current in the series resistor | A |
| `IL` | out | load current | A |
| `IZ` | out | Zener current | A |
| `IZ_mA` | out | Zener current in milliamperes | , |
| `PZ` | out | power dissipated in the Zener | W |

**Applies when:** the Zener is in breakdown (ideal: constant Vz); positive values.

```
$ electrical-engineering solve zener_regulator "Vin=20 V" "R=1000 ohm" "Vz=12 V" "RL=2000
ohm"
VL = 12 V
IR = 0.008 A
IL = 0.006 A
IZ = 0.002 A
IZ_mA = 2
PZ = 0.024 W
  by: Zener diode shunt regulator  (I_R = (Vin - Vz)/R, I_L = Vz/RL, I_Z = I_R - I_L)
  Vin = 20 V
  R = 1000 ohm
  Vz = 12 V
  RL = 2000 ohm
  assumes: the Zener is in breakdown (ideal: constant Vz); positive values
```

**What the worked example means:** This calculation determines the operating point of a Zener diode shunt regulator. The series resistor current 0.008 A A is derived from the difference between the input voltage 20 V and the Zener breakdown voltage 12 V, divided by the resistance 1000 ohm. The load current 0.006 A A flows through the load resistance 2000 ohm, resulting in a stable load voltage 12 V. The remaining current, 0.002 A A or 2 mA, passes through the Zener diode, dissipating power of 0.024 W W to maintain regulation.

## Electromagnetic induction (4)

### Induced emf from a change of flux (Faraday's law), `faraday_emf`

The average emf induced in a coil of N turns when the flux linking it changes by dPhi in a time dt, its sign by Lenz's law, and the change of flux linkage.

$$e = -N\frac{\Delta\Phi}{\Delta t}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `N` | in | number of turns | , |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `dPhi` | in | change of flux (negative for a decrease) | Wb |
| `dt` | in | time taken by the change | s |
| `emf` | out | magnitude of the average induced emf | V |
| `emf_signed` | out | induced emf with Lenz's sign, -N dPhi/dt | V |
| `flux_rate` | out | rate of change of flux in Wb/s | , |
| `linkage_change` | out | change of flux linkage N dPhi in weber-turns | , |

**Applies when:** a uniform rate of change (the result is the average emf); the flux links every turn.

```
$ electrical-engineering solve faraday_emf N=200 "dPhi=0.025 Wb" "dt=0.05 s"
emf = 100 V
emf_signed = -100 V
flux_rate = 0.5
linkage_change = 5
  by: Induced emf from a change of flux (Faraday's law)  (e = -N dPhi/dt)
  N = 200
  dPhi = 0.025 Wb
  dt = 0.05 s
  assumes: a uniform rate of change (the result is the average emf); the flux links every
turn
```

In the worked example, N = 200, dPhi = 0.025 Wb, dt = 0.05 s go through the equation above and give emf = 100 V, emf_signed = -100 V, flux_rate = 0.5, linkage_change = 5. It applies only when a uniform rate of change (the result is the average emf); the flux links every turn: outside that it says what is wrong instead of returning a number.

### Inductance of a coil on a uniform core, `coil_inductance`

The self-inductance N^2/S of a coil of N turns on a uniform core (or a long air-cored coil), with the reluctance of the flux path and the flux per ampere.

$$L = \frac{N^2}{S} = \frac{\mu_0 \mu_r N^2 A}{l}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `N` | in | number of turns | , |
| `length` | in | length of the flux path (coil length for an air-cored solenoid) | m |
| `area` | in | cross-sectional area of the core | m^2 |
| `mu_r` | in | relative permeability of the core (1 for air) | , |
| `L` | out | self-inductance | H |
| `reluctance` | out | reluctance of the flux path in A/Wb | , |
| `flux_per_ampere` | out | flux per ampere of coil current, Wb/A | , |

**Applies when:** uniform field along the path, no leakage (a long coil or a closed core); linear core.

```
$ electrical-engineering solve coil_inductance N=500 "length=0.2 m" "area=0.0004 m^2"
mu_r=1
L = 0.0006283185307 H
reluctance = 397887357.7
flux_per_ampere = 1.256637061e-06
  by: Inductance of a coil on a uniform core  (S = l/(mu0 mu_r A); L = N^2/S = mu0 mu_r N^2
A/l)
  N = 500
  length = 0.2 m
  area = 0.0004 m^2
  mu_r = 1
  assumes: uniform field along the path, no leakage (a long coil or a closed core); linear
core
```

In the worked example, N = 500, length = 0.2 m, area = 0.0004 m^2, mu_r = 1 go through the equation above and give L = 0.0006283185307 H, reluctance = 397887357.7, flux_per_ampere = 1.256637061e-06. It applies only when uniform field along the path, no leakage (a long coil or a closed core); linear core: outside that it says what is wrong instead of returning a number.

**Mutual inductance, induced emfs and coupled coils in series, `mutual_inductance`**

The mutual inductance M = k sqrt(L1 L2) of two coupled coils, the emfs induced in coil 1 and in coil 2 when the current in coil 1 changes by dI1 in dt, and the inductance of the two coils in series aiding and opposing.

$$M = k\sqrt{L_1 L_2}, \quad E_2 = M\frac{\Delta I_1}{\Delta t}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `L1` | in | self-inductance of coil 1 | H |
| `L2` | in | self-inductance of coil 2 | H |
| `k` | in | coupling coefficient (0 to 1) | , |
| `dI1` | in | change of the current in coil 1 | A |
| `dt` | in | time taken by the change | s |
| `M` | out | mutual inductance | H |
| `E1` | out | magnitude of the emf self-induced in coil 1 | V |
| `E2` | out | magnitude of the emf induced in coil 2 | V |
| `L_aiding` | out | series aiding L1 + L2 + 2M | H |
| `L_opposing` | out | series opposing L1 + L2 - 2M | H |

**Applies when:** linear coils; a uniform rate of change of current (average emfs); 0 <= k <= 1.

```
$ electrical-engineering solve mutual_inductance "L1=0.2 H" "L2=0.8 H" k=0.5 "dI1=4 A"
"dt=0.02 s"
M = 0.2 H
E1 = 40 V
E2 = 40 V
L_aiding = 1.4 H
L_opposing = 0.6 H
  by: Mutual inductance, induced emfs and coupled coils in series  (M = k sqrt(L1 L2); E1 =
L1 dI1/dt; E2 = M dI1/dt; L = L1 + L2 +/- 2M)
  L1 = 0.2 H
  L2 = 0.8 H
  k = 0.5
  dI1 = 4 A
  dt = 0.02 s
  assumes: linear coils; a uniform rate of change of current (average emfs); 0 <= k <= 1
```

In the worked example, L1 = 0.2 H, L2 = 0.8 H, k = 0.5, dI1 = 4 A, dt = 0.02 s go through the equation above and give M = 0.2 H, E1 = 40 V, E2 = 40 V, L_aiding = 1.4 H. It applies only when linear coils; a uniform rate of change of current (average emfs); 0 <= k <= 1: outside that it says what is wrong instead of returning a number.

**Self-inductance from flux linkage, `self_inductance`**

The self-inductance L = N phi/I of a coil of N turns in which a current I sets up a flux phi.

$$L = \frac{N\Phi}{I}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `N` | in | number of turns | , |
| `flux` | in | flux linking every turn | Wb |
| `I` | in | coil current | A |
| `L` | out | self-inductance | H |
| `flux_linkage` | out | flux linkage N phi in weber-turns | , |

**Applies when:** the flux links every turn; a linear core (flux proportional to the current).

```
$ electrical-engineering solve self_inductance N=200 "flux=0.005 Wb" "I=2 A"
L = 0.5 H
flux_linkage = 1
  by: Self-inductance from flux linkage  (L = N phi / I)
  N = 200
  flux = 0.005 Wb
  I = 2 A
  assumes: the flux links every turn; a linear core (flux proportional to the current)
```

In the worked example, N = 200, flux = 0.005 Wb, I = 2 A go through the equation above and give L = 0.5 H, flux_linkage = 1. It applies only when the flux links every turn; a linear core (flux proportional to the current): outside that it says what is wrong instead of returning a number.

## Electromagnetics (2)

### Plane wave in a conducting (lossy) medium: attenuation, skin depth, speed and wave impedance, `plane_wave_lossy`

The attenuation and phase constants, skin depth, phase velocity, wavelength, intrinsic impedance (magnitude and angle) and loss tangent of a uniform plane wave in a medium of conductivity sigma.

$$\gamma = \alpha + j\beta = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)}, \quad \delta = \frac{1}{\alpha}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `f` | in | frequency | Hz |
| `eps_r` | in | relative permittivity | , |
| `mu_r` | in | relative permeability | , |
| `sigma` | in | conductivity in S/m | , |
| `alpha` | out | attenuation constant in Np/m | , |
| `beta` | out | phase constant in rad/m | , |
| `skin_depth` | out | skin depth 1/alpha | m |
| `v` | out | phase velocity w/beta | m/s |
| `wavelength` | out | wavelength 2 pi/beta | m |
| `eta` | out | magnitude of the intrinsic impedance | ohm |
| `eta_angle_deg` | out | angle of the intrinsic impedance in degrees (0 to 45) | , |
| `loss_tangent` | out | loss tangent sigma/(w eps) | , |

**Applies when:** a linear, homogeneous medium; sigma > 0 (a lossless medium: plane_wave_lossless); eps_r, mu_r >= 1.

```
$ electrical-engineering solve plane_wave_lossy "f=1000000 Hz" eps_r=1 mu_r=1
sigma=58000000
alpha = 15131.91403
beta = 15131.91403
skin_depth = 6.60854931e-05 m
v = 415.2273993 m/s
wavelength = 0.0004152273993 m
eta = 0.0003689613455 ohm
eta_angle_deg = 45
loss_tangent = 1.042556007e+12
  by: Plane wave in a conducting (lossy) medium: attenuation, skin depth, speed and wave
impedance  (gamma = alpha + j beta = sqrt(jw mu (sigma + jw eps)); delta = 1/alpha; eta =
sqrt(jw mu/(sigma + jw eps)))
  f = 1000000 Hz
  eps_r = 1
  mu_r = 1
  sigma = 58000000
  assumes: a linear, homogeneous medium; sigma > 0 (a lossless medium:
plane_wave_lossless); eps_r, mu_r >= 1
```

In the worked example, f = 1000000 Hz, eps_r = 1, mu_r = 1, sigma = 58000000 go through the equation above and give alpha = 15131.91403, beta = 15131.91403, skin_depth = 6.60854931e-05 m, v = 415.2273993 m/s. It applies only when a linear, homogeneous medium; sigma > 0 (a lossless medium: plane_wave_lossless); eps_r, mu_r >= 1: outside that it says what is wrong instead of returning a number.

**Plane wave in a lossless medium: speed, wavelength and wave impedance, `plane_wave_lossless`**

The phase velocity, wavelength, intrinsic (wave) impedance, phase constant and refractive index of a uniform plane wave in a lossless medium.

$$v = \frac{c}{\sqrt{\varepsilon_r \mu_r}}, \quad \lambda = \frac{v}{f}, \quad \eta = \sqrt{\frac{\mu}{\varepsilon}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `f` | in | frequency | Hz |
| `eps_r` | in | relative permittivity | , |
| `mu_r` | in | relative permeability | , |
| `v` | out | phase velocity | m/s |
| `wavelength` | out | wavelength in the medium | m |
| `eta` | out | intrinsic impedance | ohm |
| `beta` | out | phase constant in rad/m | , |
| `n` | out | refractive index sqrt(eps_r mu_r) | , |

**Applies when:** a lossless, linear, homogeneous medium (conductivity 0); eps_r, mu_r >= 1.

```
$ electrical-engineering solve plane_wave_lossless "f=100000000 Hz" eps_r=4 mu_r=1
v = 149896229 m/s
wavelength = 1.49896229 m
eta = 188.3651567 ohm
beta = 4.191690044
n = 2
  by: Plane wave in a lossless medium: speed, wavelength and wave impedance  (v =
c/sqrt(eps_r mu_r); wavelength = v/f; eta = eta0 sqrt(mu_r/eps_r))
  f = 100000000 Hz
  eps_r = 4
  mu_r = 1
  assumes: a lossless, linear, homogeneous medium (conductivity 0); eps_r, mu_r >= 1
```

In the worked example, f = 100000000 Hz, eps_r = 4, mu_r = 1 go through the equation above and give v = 149896229 m/s, wavelength = 1.49896229 m, eta = 188.3651567 ohm, beta = 4.191690044. It applies only when a lossless, linear, homogeneous medium (conductivity 0); eps_r, mu_r >= 1: outside that it says what is wrong instead of returning a number.

## Magnetic circuits (1)

**Series magnetic circuit: reluctance, magnetomotive force, flux density and field strength, `magnetic_circuit`**

The current a coil of N turns needs to set up a flux in a series magnetic circuit of sections (iron paths and air gaps), or the flux a current produces, with the reluctance, the mmf, and B and H in every section.

$$S = \sum_k \frac{l_k}{\mu_0 \mu_{r,k} A_k}, \quad NI = \Phi S = \sum_k H_k l_k$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `N` | in | number of turns of the magnetising coil | , |
| `lengths` | in | mean length of the flux path in each section, in m (an air gap is a section) | , |
| `areas` | in | cross-sectional area of each section, in m^2 | , |
| `mu_r` | in | relative permeability of each section (1 for air) | , |
| `flux` | in | flux to produce (give this or the current) (optional) | Wb |
| `current` | in | coil current (give this or the flux) (optional) | A |
| `reluctance` | out | total reluctance in A/Wb (ampere-turns per weber) | , |
| `mmf` | out | magnetomotive force N I in ampere-turns | , |
| `current` | out | coil current | A |
| `flux` | out | flux in the circuit | Wb |
| `B` | out | flux density of each section in T | , |
| `H` | out | magnetic field strength of each section in A/m | , |

**Applies when:** linear materials: a constant relative permeability in each section (no B-H curve); no leakage or fringing; exactly one of the flux and the current.

```
$ electrical-engineering solve magnetic_circuit N=500 "flux=0.0006 Wb" "lengths=[0.4,
0.001]" "areas=[0.0005, 0.0005]" "mu_r=[800, 1]"
reluctance = 2387324.146
mmf = 1432.394488
current = 2.864788976 A
flux = 0.0006 Wb
B = [1.2,1.2]
H = [1193.6620731892149,954929.658551372]
  by: Series magnetic circuit: reluctance, magnetomotive force, flux density and field
strength  (S = sum l/(mu0 mu_r A); F = N I = flux S; B = flux/A; H = B/(mu0 mu_r))
  N = 500
  flux = 0.0006 Wb
  assumes: linear materials: a constant relative permeability in each section (no B-H
curve); no leakage or fringing; exactly one of the flux and the current
```

In the worked example, N = 500, flux = 0.0006 Wb, lengths = [0.4, 0.001], areas = [0.0005, 0.0005], mu_r = [800, 1] go through the equation above and give reluctance = 2387324.146, mmf = 1432.394488, current = 2.864788976 A, flux = 0.0006 Wb. It applies only when linear materials: a constant relative permeability in each section (no B-H curve); no leakage or fringing; exactly one of the flux and the current: outside that it says what is wrong instead of returning a number.

## Measuring instruments (2)

### Ammeter shunt resistance, `ammeter_shunt`

The shunt resistance that extends a moving-coil meter of full-scale current Ifs and resistance Rm to read a current I at full scale.

$$R_{sh} = \frac{I_{fs} R_m}{I - I_{fs}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Ifs` | in | full-scale deflection current of the meter | A |
| `Rm` | in | resistance of the meter | ohm |
| `I` | in | required full-scale reading of the ammeter | A |
| `Rsh` | out | shunt resistance | ohm |
| `multiplying_factor` | out | range multiplication I/Ifs | , |
| `V_fs` | out | voltage across the meter at full scale | V |
| `R_total` | out | resistance of the extended ammeter | ohm |

**Applies when:** I > Ifs; the meter's resistance constant.

```
$ electrical-engineering solve ammeter_shunt "Ifs=0.001 A" "Rm=100 ohm" "I=1 A"
Rsh = 0.1001001001 ohm
multiplying_factor = 1000
V_fs = 0.1 V
R_total = 0.1 ohm
  by: Ammeter shunt resistance  (Rsh = Ifs Rm / (I - Ifs))
  Ifs = 0.001 A
  Rm = 100 ohm
  I = 1 A
  assumes: I > Ifs; the meter's resistance constant
```

In the worked example, Ifs = 0.001 A, Rm = 100 ohm, I = 1 A go through the equation above and give Rsh = 0.1001001001 ohm, multiplying_factor = 1000, V_fs = 0.1 V, R_total = 0.1 ohm. It applies only when I > Ifs; the meter's resistance constant: outside that it says what is wrong instead of returning a number.

### Voltmeter multiplier resistance, `voltmeter_multiplier`

The series (multiplier) resistance that makes a moving-coil meter of full-scale current Ifs and resistance Rm read a voltage V at full scale, and the voltmeter's sensitivity.

$$R_{mult} = \frac{V}{I_{fs}} - R_m$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Ifs` | in | full-scale deflection current of the meter | A |
| `Rm` | in | resistance of the meter | ohm |
| `V` | in | required full-scale reading of the voltmeter | V |
| `Rmult` | out | multiplier (series) resistance | ohm |
| `R_total` | out | resistance of the voltmeter | ohm |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `sensitivity` | out | sensitivity in ohm per volt, 1/Ifs | , |
| `V_fs` | out | full-scale voltage of the bare meter | V |

**Applies when:** V > Ifs Rm.

```
$ electrical-engineering solve voltmeter_multiplier "Ifs=0.001 A" "Rm=100 ohm" "V=10 V"
Rmult = 9900 ohm
R_total = 10000 ohm
sensitivity = 1000
V_fs = 0.1 V
  by: Voltmeter multiplier resistance  (Rmult = V/Ifs - Rm)
  Ifs = 0.001 A
  Rm = 100 ohm
  V = 10 V
  assumes: V > Ifs Rm
```

In the worked example, Ifs = 0.001 A, Rm = 100 ohm, V = 10 V go through the equation above and give Rmult = 9900 ohm, R_total = 10000 ohm, sensitivity = 1000, V_fs = 0.1 V. It applies only when V > Ifs Rm: outside that it says what is wrong instead of returning a number.

## Operational amplifiers (5)

### Difference amplifier output (four-resistor op-amp circuit), `difference_amplifier`

The output of the classic difference amplifier: V1 through R1 to the inverting input, R2 from output to that input, V2 through R3 to the non-inverting input, R4 from it to ground.

$$V_{out} = \frac{R_4}{R_3 + R_4}\left(1 + \frac{R_2}{R_1}\right)V_2 - \frac{R_2}{R_1}V_1$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V1` | in | voltage at the inverting side | V |
| `V2` | in | voltage at the non-inverting side | V |
| `R1` | in | input resistor to the inverting input | ohm |
| `R2` | in | feedback resistor | ohm |
| `R3` | in | input resistor to the non-inverting input | ohm |
| `R4` | in | resistor from the non-inverting input to ground | ohm |
| `Vout` | out | output voltage | V |
| `gain_inverting` | out | gain from V1, -R2/R1 | , |
| `gain_noninverting` | out | gain from V2, R4/(R3+R4) (1 + R2/R1) | , |

**Applies when:** ideal op-amp (infinite gain, no saturation).

```
$ electrical-engineering solve difference_amplifier "R1=10000 ohm" "R2=100000 ohm"
"R3=10000 ohm" "R4=100000 ohm" "V1=1 V" "V2=1.5 V"
Vout = 5 V
gain_inverting = -10
gain_noninverting = 10
  by: Difference amplifier output (four-resistor op-amp circuit)  (superposition with v+ =
v-)
  R1 = 10000 ohm
  R2 = 100000 ohm
  R3 = 10000 ohm
  R4 = 100000 ohm
  V1 = 1 V
  V2 = 1.5 V
  assumes: ideal op-amp (infinite gain, no saturation)
```

**What the worked example means:** This calculation determines the output voltage of a four-resistor difference amplifier using superposition. The result, 5 V, combines the scaled input from the inverting side and the non-inverting side. The effective amplification for the first signal is -10, while the second signal sees a gain of 10. These values indicate how strongly each input influences the final output relative to their respective source voltages.

**Inverting amplifier gain and output, `inverting_amplifier`**

The voltage gain -Rf/Rin of an ideal inverting amplifier and its output for an input voltage.

$$A_v = -\frac{R_f}{R_{in}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Rin` | in | input resistor | ohm |
| `Rf` | in | feedback resistor | ohm |
| `Vin` | in | input voltage | V |
| `gain` | out | closed-loop voltage gain | , |
| `Vout` | out | output voltage | V |

**Applies when:** ideal op-amp, no saturation.

```
$ electrical-engineering solve inverting_amplifier "Rin=10000 ohm" "Rf=47000 ohm" "Vin=0.5
V"
gain = -4.7
Vout = -2.35 V
  by: Inverting amplifier gain and output  (gain = -Rf/Rin)
  Rin = 10000 ohm
  Rf = 47000 ohm
  Vin = 0.5 V
  assumes: ideal op-amp, no saturation
```

**What the worked example means:** This calculation determines the closed-loop voltage gain and the resulting output voltage for an ideal inverting operational amplifier. The gain is derived from the ratio of the feedback resistor to the input resistor, specifically negative Rf divided by Rin. Using the provided input

voltage, the final output is computed as the product of this gain and the input signal. For example, with an input resistor of 10000 ohm, a feedback resistor of 47000 ohm, and an input voltage of 0.5 V, the system yields a gain of -4.7 and an output voltage of -2.35 V. This configuration effectively scales and inverts the input signal.

**Inverting summing amplifier output, `summing_amplifier`**

The output of an ideal inverting summing amplifier: Vout = -Rf (V1/R1 + V2/R2 + ...).

$$V_{out} = -R_f \sum_k \frac{V_k}{R_k}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | input voltages in volts | , |
| `R` | in | input resistors in ohms (one per input) | , |
| `Rf` | in | feedback resistor | ohm |
| `Vout` | out | output voltage | V |
| `gains` | out | gain of each input, -Rf/R_k | , |

**Applies when:** ideal op-amp (infinite gain, no saturation); non-inverting input grounded.

```
$ electrical-engineering solve summing_amplifier "Rf=100000 ohm" "V=[1, 2, -0.5]" "R=
[10000, 20000, 10000]"
Vout = -15 V
gains = [-10.0,-5.0,-10.0]
  by: Inverting summing amplifier output  (virtual ground at the inverting input;
Kirchhoff's current law there)
  Rf = 100000 ohm
  assumes: ideal op-amp (infinite gain, no saturation); non-inverting input grounded
```

**What the worked example means:** This calculation determines the output voltage of an ideal inverting summing amplifier using the feedback resistor and individual input resistors. The result, -15 V, represents the negative sum of each input voltage divided by its corresponding resistance, scaled by the feedback resistor value. This configuration effectively performs analog addition where each signal contributes according to its specific gain factor defined by the ratio of the feedback resistor to the input resistor.

**Non-inverting amplifier gain and output, `noninverting_amplifier`**

The voltage gain 1 + Rf/Rg of an ideal non-inverting amplifier and its output.

$$A_v = 1 + \frac{R_f}{R_g}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Rg` | in | resistor from the inverting input to ground | ohm |
| `Rf` | in | feedback resistor | ohm |
| `Vin` | in | input voltage (at the non-inverting input) | V |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `gain` | out | closed-loop voltage gain | , |
| `Vout` | out | output voltage | V |

**Applies when:** ideal op-amp, no saturation.

```
$ electrical-engineering solve noninverting_amplifier "Rg=10000 ohm" "Rf=90000 ohm"
"Vin=0.2 V"
gain = 10
Vout = 2 V
  by: Non-inverting amplifier gain and output  (gain = 1 + Rf/Rg)
  Rg = 10000 ohm
  Rf = 90000 ohm
  Vin = 0.2 V
  assumes: ideal op-amp, no saturation
```

**What the worked example means:** This calculation determines the closed-loop voltage gain and the final output voltage for an ideal non-inverting amplifier. By using the feedback resistor value of 90000 ohm and the ground resistor value of 10000 ohm, the system computes a gain of 10. When this gain is applied to the input signal of 0.2 V, the predicted output voltage becomes 2 V. This result indicates how much the input signal is amplified without any phase inversion.

### Op-amp integrator output for a constant input, `opamp_integrator`

The output of an ideal inverting integrator (R in, C feedback) after a constant input Vin has been applied for time t, from an initial output V0.

$$V_{out}(t) = V_0 - \frac{1}{RC}\int_0^t V_{in}\,dt$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | input resistor | ohm |
| `C` | in | feedback capacitor | F |
| `Vin` | in | constant input voltage | V |
| `t` | in | integration time | s |
| `V0` | in | output voltage at t = 0 | V |
| `Vout` | out | output voltage at t | V |
| `slope` | out | output slope dVout/dt in V/s | , |

**Applies when:** ideal op-amp, no saturation; constant input.

```
$ electrical-engineering solve opamp_integrator "R=10000 ohm" "C=1e-06 F" "Vin=0.5 V"
"t=0.005 s" "V0=0 V"
Vout = -0.25 V
slope = -50
  by: Op-amp integrator output for a constant input  (Vout = V0 - Vin t/(RC))
  R = 10000 ohm
  C = 1e-06 F
  Vin = 0.5 V
  t = 0.005 s
  V0 = 0 V
  assumes: ideal op-amp, no saturation; constant input
```

**What the worked example means:** This calculation determines the final output voltage of an ideal inverting integrator after a constant input has been applied for a specific duration. Starting from an initial voltage, the circuit accumulates the input signal over time, producing a linear ramp at the output. The computed result shows that with an input resistor and feedback capacitor creating a specific time constant, the output reaches -0.25 V while changing at a rate of -50. This reflects how the capacitor charges or discharges in response to the steady input current generated by the voltage across the input resistor.

## Single-phase a.c. circuits (3)

### Coil with resistance in parallel with a capacitor: currents, resonance and dynamic resistance, `coil_capacitor_parallel`

For a coil (R in series with L) in parallel with a capacitor C across a supply: the coil, capacitor and supply currents, the phase angle and power at the supply frequency, and the resonant frequency, dynamic resistance L/(CR), current at resonance and Q factor.

$$f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}, \quad R_D = \frac{L}{CR}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | rms supply voltage | V |
| `f` | in | supply frequency | Hz |
| `R` | in | resistance of the coil | ohm |
| `L` | in | inductance of the coil | H |
| `C` | in | capacitance | F |
| `I_coil` | out | coil current | A |
| `I_C` | out | capacitor current | A |
| `I` | out | supply current | A |
| `Z` | out | impedance of the network V/I | ohm |
| `phase_deg` | out | phase angle of the supply current behind the voltage, degrees | , |
| `P` | out | real power (dissipated in the coil's resistance) | W |
| `f_r` | out | resonant frequency | Hz |
| `R_dynamic` | out | dynamic resistance L/(CR) at resonance | ohm |
| `I_resonance` | out | supply current at resonance | A |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `Q` | out | Q factor at resonance, wr L/R (current magnification) | , |

**Applies when:** R^2 < L/C (a resonance exists); sinusoidal steady state, ideal capacitor.

```
$ electrical-engineering solve coil_capacitor_parallel "R=10 ohm" "L=0.1 H" "C=1e-05 F"
"V=100 V" "f=50 Hz"
I_coil = 3.033144711 A
I_C = 0.3141592654 A
I = 2.735445524 A
Z = 36.55711624 ohm
phase_deg = 70.34691454
P = 91.99966835 W
f_r = 158.3571689 Hz
R_dynamic = 1000 ohm
I_resonance = 0.1 A
Q = 9.949874371
  by: Coil with resistance in parallel with a capacitor: currents, resonance and dynamic
resistance  (Y = 1/(R + jwL) + jwC; fr = (1/2 pi) sqrt(1/(LC) - R^2/L^2); RD = L/(CR); Q =
wr L/R)
  R = 10 ohm
  L = 0.1 H
  C = 1e-05 F
  V = 100 V
  f = 50 Hz
  assumes: R^2 < L/C (a resonance exists); sinusoidal steady state, ideal capacitor
```

In the worked example, R = 10 ohm, L = 0.1 H, C = 1e-05 F, V = 100 V, f = 50 Hz go through the equation above and give I_coil = 3.033144711 A, I_C = 0.3141592654 A, I = 2.735445524 A, Z = 36.55711624 ohm. It applies only when R^2 < L/C (a resonance exists); sinusoidal steady state, ideal capacitor: outside that it says what is wrong instead of returning a number.

### Parallel R-L, R-C or R-L-C circuit on an a.c. supply, `parallel_ac_supply`

The branch currents, supply current, impedance, phase angle, power factor and real, reactive and apparent power of R in parallel with L, with C, or with both, across a sinusoidal supply.

$$I = \sqrt{I_R^2 + (I_L - I_C)^2}, \quad \cos\phi = \frac{I_R}{I}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | rms supply voltage | V |
| `f` | in | supply frequency | Hz |
| `R` | in | resistance | ohm |
| `circuit` | in | the elements: R-L, R-C or R-L-C | , |
| `L` | in | inductance (R-L and R-L-C circuits) (optional) | H |
| `C` | in | capacitance (R-C and R-L-C circuits) (optional) | F |
| `IR` | out | current in R | A |
| `IL` | out | current in L (0 without an inductor) | A |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `IC` | out | current in C (0 without a capacitor) | A |
| `I` | out | supply current | A |
| `Z` | out | impedance V/I | ohm |
| `phase_deg` | out | phase angle of the supply current behind the voltage, degrees (negative: leading) | , |
| `power_factor` | out | power factor cos(phase) | , |
| `P` | out | real power in W | W |
| `Q` | out | reactive power in var (positive: inductive) | , |
| `S` | out | apparent power in VA | , |
| `nature` | out | lagging, leading or unity power factor | , |

**Applies when:** sinusoidal steady state, rms values; ideal L and C (a coil with resistance: use coil_capacitor_parallel); R > 0.

```
$ electrical-engineering solve parallel_ac_supply "R=25 ohm" "L=0.1 H" "V=100 V" "f=50 Hz"
circuit=RL
IR = 4 A
IL = 3.183098862 A
IC = 0 A
I = 5.111958369 A
Z = 19.56197465 ohm
phase_deg = 38.51188725
power_factor = 0.7824789858
P = 400 W
Q = 318.3098862
S = 511.1958369
nature = lagging
  by: Parallel R-L, R-C or R-L-C circuit on an a.c. supply  (IR = V/R, IL = V/(wL), IC = V
wC; I = sqrt(IR^2 + (IL - IC)^2); phase = atan((IL - IC)/IR))
  R = 25 ohm
  L = 0.1 H
  V = 100 V
  f = 50 Hz
  circuit = RL
  assumes: sinusoidal steady state, rms values; ideal L and C (a coil with resistance: use
coil_capacitor_parallel); R > 0
```

In the worked example, R = 25 ohm, L = 0.1 H, V = 100 V, f = 50 Hz, circuit = RL go through the equation above and give IR = 4 A, IL = 3.183098862 A, IC = 0 A, I = 5.111958369 A. It applies only when sinusoidal steady state, rms values; ideal L and C (a coil with resistance: use coil_capacitor_parallel); R > 0: outside that it says what is wrong instead of returning a number.

### Series R-L, R-C or R-L-C circuit on an a.c. supply, `series_ac_supply`

The impedance, current, voltage across each element, phase angle, power factor and real, reactive and apparent power of a series R-L, R-C or R-L-C circuit connected to a sinusoidal supply.

$$Z = \sqrt{R^2 + (X_L - X_C)^2}, \quad I = \frac{V}{Z}, \quad \cos\phi = \frac{R}{Z}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V` | in | rms supply voltage | V |
| `f` | in | supply frequency | Hz |
| `R` | in | resistance | ohm |
| `circuit` | in | the elements: R-L, R-C or R-L-C | , |
| `L` | in | inductance (R-L and R-L-C circuits) (optional) | H |
| `C` | in | capacitance (R-C and R-L-C circuits) (optional) | F |
| `Z` | out | impedance \|Z\| | ohm |
| `I` | out | rms current | A |
| `VR` | out | voltage across R | V |
| `VL` | out | voltage across L (0 without an inductor) | V |
| `VC` | out | voltage across C (0 without a capacitor) | V |
| `XL` | out | inductive reactance wL | ohm |
| `XC` | out | capacitive reactance 1/(wC) | ohm |
| `phase_deg` | out | phase angle of the supply current behind the voltage, degrees (negative: leading) | , |
| `power_factor` | out | power factor cos(phase) | , |
| `P` | out | real power in W | W |
| `Q` | out | reactive power in var (positive: inductive) | , |
| `S` | out | apparent power in VA | , |
| `nature` | out | lagging, leading or unity power factor | , |

**Applies when:** sinusoidal steady state, rms values; ideal L and C; R > 0.

```
$ electrical-engineering solve series_ac_supply "R=30 ohm" "L=0.1 H" "V=240 V" "f=50 Hz"
circuit=RL
Z = 43.43915791 ohm
I = 5.52496898 A
VR = 165.7490694 V
VL = 173.5720196 V
VC = 0 V
XL = 31.41592654 ohm
XC = 0 ohm
phase_deg = 46.32070377
power_factor = 0.6906211226
P = 915.758467 W
Q = 958.9800242
S = 1325.992555
nature = lagging
  by: Series R-L, R-C or R-L-C circuit on an a.c. supply  (Z = sqrt(R^2 + (XL - XC)^2); I =
V/Z; VR = IR, VL = I XL, VC = I XC; phase = atan((XL - XC)/R); P = I^2 R)
  R = 30 ohm
  L = 0.1 H
  V = 240 V
  f = 50 Hz
  circuit = RL
  assumes: sinusoidal steady state, rms values; ideal L and C; R > 0
```

In the worked example, R = 30 ohm, L = 0.1 H, V = 240 V, f = 50 Hz, circuit = RL go through the equation above and give Z = 43.43915791 ohm, I = 5.52496898 A, VR = 165.7490694 V, VL = 173.5720196 V. It applies only when sinusoidal steady state, rms values; ideal L and C; R > 0: outside that it says what is wrong instead of returning a number.

## Three-phase systems (1)

### Three-phase power by the two-wattmeter method, `two_wattmeter`

The total power, the power factor, the load angle and the reactive and apparent power of a balanced three-phase load from the readings of two wattmeters.

$$P = W_1 + W_2, \quad \tan\phi = \sqrt{3}\,\frac{W_1 - W_2}{W_1 + W_2}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `W1` | in | reading of the first wattmeter (negative if it reads backwards) | W |
| `W2` | in | reading of the second wattmeter | W |
| `P` | out | total power W1 + W2 | W |
| `Q` | out | total reactive power in var (magnitude) | , |
| `S` | out | total apparent power in VA | , |
| `power_factor` | out | power factor of the load | , |
| `phase_deg` | out | load angle in degrees (lag or lead needs the phase sequence) | , |

**Applies when:** a balanced load and a symmetrical supply; W1 + W2 > 0.

```
$ electrical-engineering solve two_wattmeter "W1=3000 W" "W2=1000 W"
P = 4000 W
Q = 3464.101615
S = 5291.502622
power_factor = 0.755928946
phase_deg = 40.89339465
  by: Three-phase power by the two-wattmeter method  (P = W1 + W2; tan(phi) = sqrt(3) (W1 -
W2)/(W1 + W2))
  W1 = 3000 W
  W2 = 1000 W
  assumes: a balanced load and a symmetrical supply; W1 + W2 > 0
```

In the worked example, W1 = 3000 W, W2 = 1000 W go through the equation above and give P = 4000 W, Q = 3464.101615, S = 5291.502622, power_factor = 0.755928946. It applies only when a balanced load and a symmetrical supply; W1 + W2 > 0: outside that it says what is wrong instead of returning a number.

## Transients (3)

### First-order (RC or RL) transient response, `first_order_transient`

The value at time t of any voltage in a first-order circuit going from its initial value V0 toward its final value Vf with time constant tau.

$$v(t) = V_f + (V_0 - V_f)e^{-t/\tau}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V0` | in | initial value (at t = 0+) | V |
| `Vf` | in | final (steady-state) value | V |
| `tau` | in | time constant (RC or L/R) | s |
| `t` | in | time after the switching | s |
| `v` | out | value at time t | V |
| `fraction_complete` | out | fraction of the change completed, 1 - exp(-t/tau) | , |

**Applies when:** a single time constant; a step change at t = 0.

```
$ electrical-engineering solve first_order_transient "V0=0 V" "Vf=10 V" "tau=0.001 s"
"t=0.002 s"
v = 8.646647168 V
fraction_complete = 0.8646647168
  by: First-order (RC or RL) transient response  (v(t) = Vf + (V0 - Vf) exp(-t/tau))
  V0 = 0 V
  Vf = 10 V
  tau = 0.001 s
  t = 0.002 s
  assumes: a single time constant; a step change at t = 0
```

**What the worked example means:** This calculation determines the instantaneous voltage in a first-order circuit during its transient response. Using the initial value 0 V, the final steady-state value 10 V, and the time constant 0.001 s, it computes the voltage at the specific time 0.002 s. The result, 8.64665 V, indicates the potential difference across the component at that moment. Additionally, the fraction of the transition completed is 0.864665, showing how close the system is to reaching its final state relative to the start point.

### Step response of a series RLC circuit, `series_rlc_step`

Capacitor voltage and loop current at time t after a DC step V is applied to a series RLC circuit at rest, with the damping classification.

$$LC\,\ddot{v}_C + RC\,\dot{v}_C + v_C = V, \quad \alpha = \frac{R}{2L},\ \omega_0 = \frac{1}{\sqrt{LC}}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `R` | in | resistance | ohm |
| `L` | in | inductance | H |
| `C` | in | capacitance | F |
| `V` | in | step voltage applied at t = 0 | V |
| `t` | in | time after the step | s |
| `vC` | out | capacitor voltage at t | V |
| `i` | out | loop current at t | A |

| symbol | | meaning | SI unit |
|---|---|---|---|
| `i_mA` | out | loop current in milliamperes | , |
| `alpha` | out | neper frequency R/(2L) in 1/s | , |
| `omega0` | out | undamped resonant frequency 1/sqrt(LC) | rad/s |
| `zeta` | out | damping ratio alpha/omega0 | , |
| `response` | out | underdamped, critically damped or overdamped | , |

**Applies when:** circuit at rest before the step (vC = 0, i = 0); ideal components.

```
$ electrical-engineering solve series_rlc_step "R=10 ohm" "L=0.1 H" "C=0.0001 F" "V=12 V"
"t=0.005 s"
vC = 10.41435345 V
i = 0.2992853965 A
i_mA = 299.2853965
alpha = 50
omega0 = 316.227766 rad/s
zeta = 0.158113883
response = underdamped
  by: Step response of a series RLC circuit  (closed-form solution of L C vC'' + R C vC' +
vC = V by the roots of the characteristic equation)
  R = 10 ohm
  L = 0.1 H
  C = 0.0001 F
  V = 12 V
  t = 0.005 s
  assumes: circuit at rest before the step (vC = 0, i = 0); ideal components
```

**What the worked example means:** This calculation determines the transient behavior of a series RLC circuit after a DC step voltage is applied. Using the component values, it computes the neper frequency as 50 per second and the undamped resonant frequency as 316.228 rad/s radians per second. The resulting damping ratio is 0.158114. Based on these parameters, the system exhibits an underdamped response. At time 0.005 s, the capacitor voltage reaches 10.4144 V volts and the loop current is 0.299285 A amperes, which corresponds to 299.285 milliamperes.

### Time for a first-order response to reach a value, `first_order_time_to_value`

How long a first-order (RC or RL) response from V0 toward Vf with time constant tau takes to reach a value v.

$$t = \tau \ln \frac{V_f - V_0}{V_f - v}$$

| symbol | | meaning | SI unit |
|---|---|---|---|
| `V0` | in | initial value | V |
| `Vf` | in | final value | V |
| `tau` | in | time constant | s |
| `v` | in | value to reach | V |
| `t` | out | time to reach v | s |
| `time_constants` | out | the same time in time constants | , |

**Applies when:** v between V0 and Vf.

```
$ electrical-engineering solve first_order_time_to_value "V0=0 V" "Vf=10 V" "tau=0.002 s"
"v=5 V"
t = 0.001386294361 s
time_constants = 0.6931471806
  by: Time for a first-order response to reach a value  (t = -tau ln(1 - (v - V0)/(Vf -
V0)))
  V0 = 0 V
  Vf = 10 V
  tau = 0.002 s
  v = 5 V
  assumes: v between V0 and Vf
```

**What the worked example means:** This calculation finds the elapsed time for a first-order response starting at 0 V and ending at 10 V with a time constant of 0.002 s to reach the target value 5 V. The result is 0.00138629 s, meaning the circuit takes this duration to achieve that specific level. Expressed in normalized units, this corresponds to 0.693147 time constants, indicating how far along the exponential curve the target lies.

Questions and reports: support@ai999x.com. AI999X, The Future Is Self-Improving.